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AEROSOL-GENERATING DEVICE COMPRISING AIRFLOW GUIDING ELEMENT EXTENDING INTO HEATING CHAMBER

Abstract

An aerosol-generating system is provided, including: an aerosol-generating device having a mouth end; and an aerosol-generating article, the device including: a heating chamber to receive and heat the article, and an airflow guiding element within the chamber and extending longitudinally towards the mouth end, the element to be inserted into the article upon receipt within the chamber, a width or external diameter of the element varying along a length thereof, and a first portion of the element having a width or external diameter corresponding to a maximum width or external diameter of the element, and a second portion of the element located upstream of the first portion having a width or external diameter smaller than the maximum width or external diameter, the article including: an aerosol-generating substrate as a hollow tubular segment defining a substrate cavity extending from an upstream end to a downstream end of the substrate.

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Background/Summary

[0001] The present invention relates to an aerosol-generating device for heating an aerosol-generating article comprising a rod of aerosol-generating substrate that is adapted to produce an inhalable aerosol upon heating. The present disclosure also relates to an aerosol-generating system comprising such an aerosol-generating device and an aerosol-generating article configured to be received by the aerosol-generating device.

[0002] Aerosol-generating articles in which an aerosol-generating substrate, such as a tobacco-containing substrate, is heated rather than combusted, are known in the art. Typically, in such heated smoking articles an aerosol is generated by the transfer of heat from a heat source to a physically separate aerosol-generating substrate or material, which may be located in contact with, within, around, or downstream of the heat source. During use of the aerosol-generating article, volatile compounds are released from the aerosol-generating substrate by heat transfer from the heat source and are entrained in air drawn through the aerosol-generating article. As the released compounds cool, they condense to form an aerosol.

[0003] A number of prior art documents disclose aerosol-generating devices for consuming aerosol-generating articles. Such devices include, for example, electrically heated aerosol-generating devices in which an aerosol is generated by the transfer of heat from one or more electrical heater elements of the aerosol-generating device to the aerosol-generating substrate of a heated aerosol-generating article. For example, electrically heated aerosol-generating devices have been proposed that comprise an internal heater blade which is adapted to be inserted into the aerosol-generating substrate. Use of an aerosol-generating article in combination with an external heating system is also known. For example, WO-A-2020/115151 describes the provision of an external heating element arranged around the periphery of the aerosol-generating article when the aerosol-generating article is received in a cavity of the aerosol-generating device. As an alternative, inductively heatable aerosol-generating articles comprising an aerosol-generating substrate and a susceptor arranged within the aerosol-generating substrate have been proposed by WO-A-2015/176898.

[0004] In general, it can be difficult to provide efficient heating of an aerosol-generating substrate throughout the whole rod of the substrate. The portions of the substrate closest to the heating element will inevitably be heated most effectively whilst the imperfect transfer of heat through the substrate will mean that portions of the substrate furthest from the heating element may not be effectively heated. The generation of aerosol from these portions of the substrate that are not effectively heated is therefore not optimal and, in some cases, parts of the substrate may not reach a sufficiently high temperature during use for an aerosol to be generated at all. For example, where an external heating element is used to heat a rod of aerosol-generating substrate, as described above, the central portion of the rod of aerosol-generating substrate is unlikely to generate as much aerosol as the outer portions of the rod and in some cases, may not generate any aerosol. Overall, the generation of aerosol from the aerosol-generating rod is therefore likely to be inefficient, with potential waste of a portion of the aerosol-generating substrate.

[0005] In addition, aerosol is generally not immediately generated by the aerosol-generating substrate upon activation of a heating element. This is because there is a pre-heating time after

activation of a heating element during which the aerosol-generating substrate is heated to a temperature required for aerosol generation. As such, there may be a relatively long duration between activation of a heating element and generation of a sensorially acceptable aerosol for inhalation by a user.

[0006] It would therefore be desirable to provide an aerosol-generating article having an aerosol-generating substrate that is adapted to provide more efficient aerosolisation of the aerosol-generating substrate and that reduces waste of the substrate materials, such as tobacco. It would also be desirable to provide such an aerosol-generating article that can achieve a relatively short pre-heating time so that a sensorially acceptable aerosol can be delivered to a user shortly after initiation of heating of the aerosol-generating substrate. It would also be desirable to provide such an aerosol-generating article that can provide optimised delivery of aerosol from the aerosol-generating substrate. It would be particularly desirable to provide such an aerosol-generating article with a relatively simple design so that it can be manufactured in a cost-effective way and incorporated into existing product designs. It would be further desirable to provide such an article that can be readily adapted so that it can be heated in a variety of types of heating device, including inductive and resistive heating devices.

[0007] There is provided an aerosol-generating device having a mouth end for receiving an aerosol-generating article. The aerosol-generating device may comprise a heating chamber for receiving an aerosol-generating article at the mouth end of the device. The aerosol-generating device may comprise a heating chamber for receiving and heating an aerosol-generating article. The aerosol-generating device may comprise an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the device. The airflow guiding element may comprise an elongate body. The airflow guiding element may be inserted into, or received within, an aerosol-generating article upon receipt of the article within the heating chamber. The width or external diameter of the airflow guiding element may vary along its length. A first portion of the airflow guiding element may have a width or external diameter corresponding to the maximum width or external diameter of the airflow guiding element. A second portion of the airflow guiding element located upstream of said first portion may have a width or external diameter smaller than the maximum width or external diameter of the airflow guiding element.

[0008] There is also provided an aerosol-generating device having a mouth end for receiving an aerosol-generating article. The aerosol-generating device may comprise a heating chamber for receiving an aerosol-generating article at the mouth end of the device. The aerosol-generating device may comprise a heating element provided at or about the periphery of the heating chamber for heating an aerosol-generating article received within the heating chamber. The aerosol-generating device may comprise an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the device. The airflow guiding element may be configured to be inserted into an aerosol-generating article upon receipt within the heating chamber. In other words, the airflow guiding element may be received within an aerosol-generating article when the article is received within the heating chamber.

[0009] According to the present invention, there is provided an aerosol-generating device having a mouth end for receiving an aerosol-generating article. The aerosol-generating device comprises a heating chamber for receiving and heating an aerosol-generating article. The aerosol-generating device comprises an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the device. The airflow guiding element is inserted into, or received within, an aerosol-generating article upon receipt of the article within the heating chamber. The width or external diameter of the airflow guiding element varies along its length. A first portion of the airflow guiding element has a width or external diameter corresponding to the maximum width or external diameter of the airflow guiding element. A second portion of the airflow guiding element located upstream of said first portion has a width or external diameter smaller than the maximum width or external diameter of the airflow guiding element.

[0010] There is also provided an aerosol-generating system comprising an aerosol-generating device according to the present disclosure and an aerosol-generating article for producing an inhalable aerosol upon heating. The aerosol-generating article may comprise an aerosol-generating substrate. The aerosol-generating substrate may be in the form of a hollow tubular segment defining a substrate cavity extending from an upstream end of the aerosol-generating substrate to the downstream end of the aerosol-generating substrate. Upon receipt of the aerosol-generating article within the heating chamber of the device, the airflow guiding element may extend or protrude longitudinally into the substrate cavity. As a result, an airflow channel may be defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate.

[0011] According to the present invention, there is also provided an aerosol-generating system comprising an aerosol-generating device according to the present disclosure and an aerosol-generating article for producing an inhalable aerosol upon heating. The aerosol-generating article comprises an aerosol-generating substrate. The aerosol-generating substrate is in the form of a hollow tubular segment defining a substrate cavity extending from an upstream end of the aerosol-generating substrate to the downstream end of the aerosol-generating substrate. Upon receipt of the aerosol-generating article within the heating chamber of the device, the airflow guiding element extends longitudinally into the substrate cavity. An airflow channel is defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate.

[0012] As used herein with reference to the present disclosure, the term “aerosol-generating article” is used to describe an article comprising an aerosol-generating substrate that is heated to generate an inhalable aerosol for delivery to a user.

[0013] As used herein with reference to the present disclosure, the term “aerosol-generating substrate” is used to describe a substrate comprising aerosol-generating material that is capable of releasing upon heating volatile compounds that can generate an aerosol.

[0014] As used herein with reference to the present disclosure, the term “aerosol” is used to describe a dispersion of solid particles, or liquid droplets, or a combination of solid particles and liquid droplets, in a gas. The aerosol may be visible or invisible. The aerosol may include vapours of substances that are ordinarily liquid or solid at room temperature as well as solid particles, or liquid droplets, or a combination of solid particles and liquid droplets.

[0015] Aerosol-generating articles according to the present disclosure have a downstream end through which, in use, an aerosol exits the aerosol-generating article for delivery to a user. The downstream end of the aerosol-generating article may also be referred to as the proximal end or mouth end of the aerosol-generating article. In use, a user draws directly or indirectly on the downstream end of the aerosol-generating article to inhale an aerosol generated by the aerosol-generating article.

[0016] Aerosol-generating articles according to the present disclosure have an upstream end. The upstream end is opposite the downstream end. The upstream end of the aerosol-generating article may also be referred to as the distal end of the aerosol-generating article.

[0017] Components or portions of aerosol-generating articles and devices according to the present disclosure may be described as being upstream or downstream of one another based on their relative positions between the upstream end of the aerosol-generating article or device and the downstream end of the aerosol-generating article or device. The downstream end of an aerosol-generating device may correspond to the mouth or distal end of the aerosol-generating device. The upstream end of an aerosol-generating device may correspond to the end of the aerosol-generating device opposite to the downstream end.

[0018] As used herein with reference to the present disclosure, the term “longitudinal” refers to the direction between the upstream end and the opposed downstream end of the aerosol-generating article or device.

[0019] As used herein with reference to the present disclosure, the term “transverse” is used to

describe the direction perpendicular to the longitudinal direction.

[0020] As used herein with reference to the present disclosure, the term “cross-section” is used to refer to the transverse cross-section of the aerosol-generating article, the aerosol-generating article, or component thereof unless stated otherwise.

[0021] As used herein with reference to the present disclosure, the term “radial” is used to describe a direction identified by a line extending in a plane perpendicular to the central longitudinal axis of the aerosol-generating article and passing through the point at which the central longitudinal axis intersects the perpendicular plane. Thus, as used herein with reference to the present disclosure, the term “radial direction” refers to a direction perpendicular to the central longitudinal axis and is used, for example, when describing an aerosol-generating article having a substantially cylindrical shape.

[0022] As used herein with reference to the present disclosure, the terms “hollow tubular element” and “hollow tubular substrate element” denote a generally elongate element defining a lumen, a cavity, or an airflow passage along a longitudinal axis thereof. In particular, the term “tubular” is used with reference to a tubular element having a substantially cylindrical cross-section and defining at least one airflow conduit establishing an uninterrupted fluid communication between an upstream end of the tubular element and a downstream end of the tubular element. However, it will be understood that alternative geometries (for example, alternative cross-sectional shapes) of the tubular element may be possible. The hollow tubular element may be an individual, discrete element of the aerosol-generating article which has a defined length and thickness. The terms “hollow tubular substrate” or “hollow tubular substrate element” refer to an aerosol-generating substrate in the form of a hollow tube.

[0023] As used herein with reference to the present disclosure, the term “homogenised tobacco material” encompasses any material formed by the agglomeration of tobacco particles. The homogenised tobacco material may be produced by casting, extrusion, paper making processes, or any other suitable processes known in the art.

[0024] As used herein with reference to the present disclosure, the term “tobacco particles” describes particles of any plant member of the genus *Nicotiana*. The term “tobacco particles” encompasses ground or powdered tobacco leaf lamina, ground or powdered tobacco leaf stems, tobacco dust, tobacco fines, and other particulate tobacco by-products formed during the treating, handling, and shipping of tobacco. Preferably, the tobacco particles are substantially all derived from tobacco leaf lamina. By contrast, isolated nicotine and nicotine salts are compounds derived from tobacco but are not considered tobacco particles for purposes of the present disclosure.

[0025] The provision of an elongate airflow guiding element extending or protruding into the heating chamber of the aerosol-generating device provides a centering means that ensures aerosol-generating articles being inserted into the device, in particular those having an open and hollow upstream end and a hollow aerosol-generating substrate, are substantially aligned with the central longitudinal axis of the heating chamber. This may be particularly important for devices having a heater located at or around the periphery of the heating chamber so as to externally heat a received aerosol-generating article. The airflow guiding element may assist in centering the aerosol-generating article and its substrate with respect to the heater of the device so that the entire periphery of the aerosol-generating substrate is substantially positioned at a uniform distance to the surrounding heater.

[0026] In aerosol-generating articles and systems according to the present disclosure, the provision of a substrate element in a tubular form may advantageously enable the amount of tobacco material in the aerosol-generating substrate to be optimised so that aerosol can be efficiently generated from the aerosol-generating substrate upon heating. The tubular form also removes a central portion of homogenised tobacco material that would potentially not be heated as effectively as an outer portion, in particular, in an aerosol-generating device comprising external heating means. Overall, the amount of tobacco material can therefore be significantly reduced compared to conventional

solid plugs of homogenised tobacco material and tobacco waste can be reduced. For example, it has been found that the amount of tobacco material used in the hollow tubular substrate element of aerosol-generating articles according to the present disclosure can be reduced by up to 40 percent compared to the amount of tobacco material used in the solid plug of substrate in a conventional aerosol-generating article, whilst retaining a similar delivery of aerosol to the consumer.

[0027] The amount of tobacco material provided in the substrate can be readily adapted through controlling the parameters of the hollow tubular substrate element, such as the density of a peripheral wall of the hollow tubular substrate element and the wall thickness. In this way, it is possible to adapt the hollow tubular substrate element so that it matches the heating zone of an associated aerosol-generating device. The proportion of the aerosol-generating substrate that can be heated to the necessary temperature for aerosol generation is therefore maximised so that the generation of aerosol from the aerosol-generating substrate is optimised.

[0028] The hollow tubular substrate element has a relatively simple structure that can be produced in a straightforward and cost-effective way, using existing apparatus. The hollow tubular substrate element can then be incorporated into aerosol-generating articles with other components, using known assembly methods and apparatus.

[0029] When an aerosol-generating article comprising an open upstream end and a hollow tubular aerosol-generating substrate is received within an aerosol-generating device comprising an airflow guiding element extending longitudinally into and along the heating chamber, the article can receive the airflow guiding element of the device via the upstream end of the article so that the airflow guiding element extends longitudinally into an empty substrate cavity defined by a hollow tubular aerosol-generating substrate. As a result, an airflow channel or path is defined around the airflow guiding element, between an external surface of the airflow guiding element and the interior of the aerosol-generating substrate so that air entering the substrate cavity is encouraged to flow closer to the inner surface of the aerosol-generating substrate. The protruding of the airflow guiding element into the cavity effectively provides an airflow path having a reduced cross-section relative to the cross-section of the cavity such that air entering the cavity may be locally accelerated when flowing through such an airflow path, in line with Bernoulli's principle. Such local airflow acceleration may enhance the extraction of aerosol-forming components from the heated aerosol-generating substrate, which is particularly important for a tubular substrate having an empty core. As a result, this airflow arrangement may improve aerosol delivery arising from articles having a hollow tubular substrate and reduce the time required for the substrate to generate sensorially acceptable aerosol after initially being heated, while reducing manufacturing costs and potential waste of substrate material that may not have contributed to aerosol generation.

[0030] Furthermore, the provision of an upstream portion of the airflow guiding element having a diameter smaller than a downstream portion of the airflow guiding element having a diameter corresponding to the maximum diameter of the airflow guiding element ensures that the air flowing around the airflow guiding element encounters a decrease in the cross-sectional area of the airflow channel defined between the external surface of the airflow guiding element and the internal surface of a hollow aerosol-generating substrate of an aerosol-generating article received in the device cavity of the aerosol-generating device, as a result of the increase in diameter of the airflow guiding element, further along the airflow guiding element and not at the base (or upstream end or proximal end) of the airflow guiding element. This diameter or width increase may disturb the air flow and push the air closer to the internal surface of a received hollow aerosol-generating substrate, thereby encouraging localised turbulent airflow and localised air flow acceleration to form between the airflow guiding element, particularly at the location where the diameter or width increases, and an inner surface of the hollow aerosol-generating substrate. This may improve aerosol generation as a result of the enhanced mixing of the air with aerosol-forming components released from aerosol-generating substrate upon heating.

[0031] As mentioned above, aerosol-generating articles according to the present disclosure

comprise a hollow tubular substrate element. Preferably, the hollow tubular substrate element is formed of homogenised tobacco material. Preferably, the hollow tubular substrate element is formed of one or more layers of homogenised tobacco material, such as cast leaf.

[0032] Preferably, the hollow tubular substrate element is formed of 2 or more overlapping layers of homogenised tobacco material, more preferably 3 or more overlapping layers of homogenised tobacco material.

[0033] The hollow tubular substrate element is preferably formed of up to 10 overlapping layers of homogenised tobacco material, more preferably up to 5 overlapping layers of homogenised tobacco material. For example, the hollow tubular substrate element may be formed of between about 2 and about 10 overlapping layers of homogenised tobacco material, or between about 3 and about 5 overlapping layers of homogenised tobacco material.

[0034] Preferably, the plurality of overlapping layers of homogenised tobacco material are directly overlying each other so that adjacent layers are in direct contact with each other, without intermediate layers.

[0035] The multi-layered arrangement of the layers may provide a relatively dense structure which has sufficient structural rigidity to provide the aerosol-generating substrate in an aerosol-generating article without the need for any additional support, such as carrier layers or internal support members within the longitudinal substrate cavity.

[0036] Preferably the layers of homogenised tobacco material are in sheet form. As used herein with reference to the present disclosure, the term “sheet” describes a laminar element having a width and length substantially greater than the thickness thereof.

[0037] The hollow tubular substrate element may have a length of at least about 5 millimetres, or at least about 7 millimetres, or at least about 10 millimetres.

[0038] The hollow tubular substrate element may have a length of up to about 30 millimetres, up to about 25 millimetres, or up to about 20 millimetres.

[0039] For example, the hollow tubular substrate element may have a length of between about 5 millimetres and about 30 millimetres, or between about 7 millimetres and about 25 millimetres, or between about 10 millimetres and about 20 millimetres.

[0040] Preferably, the hollow tubular substrate element has a length of about 12 millimetres.

[0041] As discussed above, the length of the hollow tubular substrate element may advantageously be matched to the longitudinal dimensions of the heating element in the corresponding aerosol-generating device which will be used to heat the aerosol-generating article. In this way, as much as possible of the aerosol-generating substrate can be heated during use, in order to optimise the amount of aerosol that can be generated and reduce the amount of tobacco waste.

[0042] Preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is at least about 0.1. More preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is at least about 0.15. More preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is at least about 0.2.

[0043] Preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is up to about 0.6. More preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is up to about 0.55. More preferably, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article is up to about 0.5.

[0044] For example, the ratio of the length of the hollow tubular substrate element to the overall length of the aerosol-generating article may be between about 0.1 and about 0.6, more preferably between about 0.15 and about 0.55, more preferably between about 0.2 and about 0.5.

[0045] Preferably, the hollow tubular substrate element has an external diameter less than an external diameter of the aerosol-generating article.

[0046] Preferably, the hollow tubular substrate element may have an external diameter of at least

about 5 millimetres, or at least about 5.5 millimetres, or at least 6 millimetres.

[0047] Preferably, the hollow tubular substrate element may have an external diameter of up to about 9 millimetres, or up to about 8 millimetres, or up to about 7.5 millimetres.

[0048] For example, the hollow tubular substrate element may have an external diameter of between about 5 millimetres and about 9 millimetres, or between about 5.5 millimetres and 8 millimetres, or between about 6 millimetres and 7.5 millimetres.

[0049] Preferably, the external diameter of the hollow tubular substrate element is substantially constant along the length of the hollow tubular substrate. As an alternative, different portions of the hollow tubular substrate element may have different external diameters.

[0050] As used herein with reference to the present disclosure, the term “external diameter” refers to the maximum diameter of the aerosol-generating article or component thereof, in the transverse direction of the aerosol-generating article, at a position along the length of the aerosol-generating article or component thereof. Where a range or value for an external diameter of the aerosol-generating article or component thereof is described herein, the external diameter of the aerosol-generating article or component thereof along the entire length of the aerosol-generating article or component thereof may fall within the same range or have the same value. In other words, where a range or value for an external diameter of the aerosol-generating article or component thereof is described herein, the external diameter of the aerosol-generating article or component thereof at all positions along the length of the aerosol-generating article or component thereof may fall within the same range or have the same value.

[0051] The external diameter of the hollow tubular substrate element does not include the width of any other component of the aerosol-generating substrate located externally of the hollow tubular substrate element.

[0052] The hollow tubular substrate element has a peripheral wall which defines the longitudinal cavity or substrate cavity. A wall thickness of the hollow tubular substrate element may be selected based on a desired amount of tobacco material within the hollow tubular substrate. A wall thickness of the hollow tubular substrate element may also be selected such the hollow tubular substrate element has a sufficiently high rigidity that it can be self-supporting. A wall thickness of the hollow tubular substrate may also be selected such that the longitudinal cavity has a cross-sectional area that provides the hollow tubular substrate element with a desired resistance to draw (RTD).

[0053] The hollow tubular substrate element may have a wall thickness that is at least about 4 percent of an external diameter of the hollow tubular substrate element, or at least about 5 percent of an external diameter of the hollow tubular substrate element, or at least about 6 percent of an external diameter of the hollow tubular substrate element.

[0054] The hollow tubular substrate element may have a wall thickness that is up to about 40 percent of an external diameter of the hollow tubular substrate element, or up to about 30 percent of an external diameter of the hollow tubular substrate element, or up to about 20 percent of an external diameter of the hollow tubular substrate element.

[0055] For example, the hollow tubular substrate element may have a wall thickness that is between about 4 percent and about 40 percent of an external diameter of the hollow tubular substrate element, or between about 5 percent and about 30 percent of an external diameter of the hollow tubular substrate element, or between about 6 percent and about 20 percent of an external diameter of the hollow tubular substrate element.

[0056] Preferably, the hollow tubular substrate element has a wall thickness of about 7 percent of an external diameter of the hollow tubular substrate element.

[0057] The hollow tubular substrate element may have a wall thickness of at least about 0.3 millimetres, or at least about 0.35 millimetres, or at least about 0.4 millimetres. The hollow tubular substrate element may have a wall thickness of at least about 0.5 millimetres. The hollow tubular substrate element may have a wall thickness of at least about 0.6 millimetres.

[0058] The hollow tubular substrate element may have a wall thickness of at least about 0.8

millimetres. The hollow tubular substrate element may have a wall thickness of at least about 1 millimetre.

[0059] The hollow tubular substrate element may have a wall thickness of up to about 3 millimetres, or up to about 2 millimetres, or up to about 1 millimetre.

[0060] For example, the hollow tubular substrate element may have a wall thickness of between about 0.3 millimetres and about 3 millimetres, or between about 0.35 millimetres and about 2 millimetres, or between about 0.4 millimetres and about 1 millimetre.

[0061] The hollow tubular substrate element may have a wall thickness of between about 0.5 millimetres and about 2 millimetres. The hollow tubular substrate element may have a wall thickness of between about 1 millimetre and about 2 millimetres.

[0062] The hollow tubular substrate element may have a wall thickness of about 0.5 millimetres. The hollow tubular substrate element may have a wall thickness of about 1 millimetre.

[0063] As described above, the longitudinal cavity provides an unrestricted flow channel through the hollow tubular substrate element. This means that the hollow tubular substrate element provides a negligible level of resistance to draw (RTD). The term “negligible level of RTD” is used to describe an RTD of less than 1 mm H₂O per 10 millimetres of length of the hollow tubular substrate element, preferably less than 0.4 mm H₂O per 10 millimetres of length of the hollow tubular substrate element, more preferably less than 0.1 mm H₂O per 10 millimetres of length of the hollow tubular substrate element.

[0064] The longitudinal cavity should therefore be free from any components that would obstruct the flow of air in a longitudinal direction. Preferably, the longitudinal cavity is substantially empty. More preferably, the longitudinal cavity is empty.

[0065] The longitudinal cavity may also be referred to as a longitudinal airflow channel.

[0066] The longitudinal cavity extends between the ends of the hollow tubular substrate element and is preferably open at both the upstream and downstream ends. The open upstream end may provide the main air inlet for drawing air through the aerosol-generating article when the consumer puffs on the article. The longitudinal cavity may therefore provide the main passageway for the flow of air and aerosol through the article.

[0067] The aerosol-generating substrate may have a length of at least about 10 millimetres, at least about 12 millimetres, or at least about 15 millimetres.

[0068] The aerosol-generating substrate may have a length of up to about 40 millimetres, up to about 37 millimetres, or up to about 35 millimetres.

[0069] For example, the aerosol-generating substrate may have a length of between about 10 millimetres and about 40 millimetres, or between about 12 millimetres and about 37 millimetres, or between about 15 millimetres and about 35 millimetres.

[0070] The diameter of the longitudinal cavity corresponds to the internal diameter of the hollow tubular substrate element.

[0071] The longitudinal cavity may have a diameter of at least about 1 millimetre, or at least about 2 millimetres, or at least about 3 millimetres.

[0072] The longitudinal cavity may have a diameter of up to about 8 millimetres, or up to about 7 millimetres, or up to about 6.5 millimetres.

[0073] For example, the longitudinal cavity may have a diameter of between about 1 millimetre and about 8 millimetres, or between about 2 millimetres and about 7 millimetres, or between about 3 millimetres and about 6.5 millimetres.

[0074] The longitudinal cavity may have a diameter of about 6 millimetres.

[0075] The diameter of the longitudinal cavity may be selected so that the volume of the cavity is sufficiently large that it provides a desired level of airflow, whilst also retaining a sufficient wall thickness. This is necessary so that there is a sufficient amount of tobacco material provided within the hollow tubular substrate element and so that the hollow tubular substrate element has a sufficiently high rigidity that it can be self-supporting.

[0076] Preferably, the longitudinal cavity has a substantially constant cross-sectional shape and size along the length of the hollow tubular substrate. However, one or both of the cross-sectional shape and size of the longitudinal cavity may vary along the length of the hollow tubular substrate element.

[0077] Preferably, the longitudinal cavity has a transverse cross-section that is substantially circular. Alternatively, the longitudinal cavity may have a transverse cross-section that is substantially oval.

[0078] The longitudinal cavity may have a constant diameter along the length of the hollow tubular substrate element. However, the diameter of the longitudinal cavity may vary along the length of the hollow tubular substrate element.

[0079] The central longitudinal axis of the hollow tubular substrate element is preferably aligned with the central longitudinal axis of other elements of the aerosol-generating article, for example other components of the aerosol-generating substrate and components of the downstream section. For example, the central longitudinal axis of the hollow tubular substrate element is preferably aligned with the central longitudinal axis of both the upstream element and the downstream element. The central longitudinal axis of the hollow tubular substrate element is preferably aligned with the central longitudinal axis of the aerosol-generating article.

[0080] The hollow tubular substrate element may comprise one or more susceptor elements located in contact with the peripheral wall, for inductive heating of the homogenised tobacco material during use.

[0081] As used herein, the term 'susceptor element' refers to an element comprising a material that is capable of converting electromagnetic energy into heat. When a susceptor element is located in an alternating electromagnetic field, the susceptor is heated. Heating of the susceptor element may be the result of at least one of hysteresis losses and eddy currents induced in the susceptor, depending on the electrical and magnetic properties of the susceptor material.

[0082] Preferably, the hollow tubular substrate element comprises one or more susceptor elements on a surface of the peripheral wall. The hollow tubular substrate element may comprise one or more susceptor elements on the inner surface of the peripheral wall, within the longitudinal airflow channel. Alternatively or in addition, the hollow tubular substrate element may comprise one or more susceptor elements on the outer surface of the peripheral wall.

[0083] The susceptor element may comprise any suitable material. The susceptor element may be formed from any material that can be inductively heated to a temperature sufficient to release volatile compounds from the aerosol-generating substrate. Suitable materials for the elongate susceptor element include graphite, molybdenum, silicon carbide, stainless steels, niobium, aluminium, nickel, nickel containing compounds, titanium, and composites of metallic materials. Some susceptor elements comprise a metal or carbon. Advantageously the susceptor element may comprise or consist of a ferromagnetic material, for example, ferritic iron, a ferromagnetic alloy, such as ferromagnetic steel or stainless steel, ferromagnetic particles, and ferrite. A suitable susceptor element may be, or comprise, aluminium. The susceptor element preferably comprises more than about 5 percent, preferably more than about 20 percent, more preferably more than about 50 percent or more than about 90 percent of ferromagnetic or paramagnetic materials. Some elongate susceptor elements may be heated to a temperature in excess of about 250 degrees Celsius.

[0084] Aerosol-generating articles according to the present disclosure may further comprise an upstream section located upstream of the rod of aerosol-generating substrate. The upstream section is preferably located immediately upstream of the rod of aerosol-generating substrate. The upstream section preferably extends between the upstream end of the aerosol-generating article and the rod of aerosol-generating substrate. The upstream section may comprise one or more upstream elements located upstream of the rod of aerosol-generating substrate. Such one or more upstream elements are described within the present disclosure.

[0085] The aerosol-generating articles of the present disclosure preferably comprise an upstream

element located upstream of and adjacent to the aerosol-generating substrate. The upstream element advantageously prevents direct physical contact with the upstream end of the aerosol-generating substrate.

[0086] Furthermore, the presence of an upstream element helps to prevent any loss of the substrate, which may be advantageous, for example, if the substrate contains particulate plant material.

[0087] Where the aerosol-generating substrate comprises shredded tobacco, such as tobacco cut filler, the upstream section or element thereof may additionally help to prevent the loss of loose particles of tobacco from the upstream end of the article. This may be particularly important when the shredded tobacco has a relatively low density, for example.

[0088] The upstream section, or an upstream element thereof, may also additionally provide a degree of protection to the aerosol-generating substrate during storage, as it covers at least to some extent the upstream end of the aerosol-generating substrate, which may otherwise be exposed.

[0089] For aerosol-generating articles that are intended to be inserted into a cavity in an aerosol-generating device such that the aerosol-generating substrate can be externally heated within the cavity, the upstream section, or an upstream element thereof, may advantageously facilitate the insertion of the upstream end of the article into the cavity. The inclusion of the upstream element may additionally protect the end of the rod of aerosol-generating substrate during the insertion of the article into the cavity such that the risk of damage to the substrate is minimised.

[0090] The upstream section, or an upstream element thereof, may also provide an improved appearance to the upstream end of the aerosol-generating article. Furthermore, if desired, the upstream section, or an upstream element thereof, may be used to provide information on the aerosol-generating article, such as information on brand, flavour, content, or details of the aerosol-generating device that the article is intended to be used with.

[0091] An upstream element may be a porous plug element. An upstream element may be made of a porous material or may comprise a plurality of openings. This may, for example, be achieved through laser perforation. Preferably, the plurality of openings is distributed homogeneously over the cross-section of the upstream element. The porosity or permeability of an upstream element may advantageously be designed in order to provide an aerosol-generating article with a particular overall resistance to draw (RTD) without substantially impacting the filtration provided by other portions of the article.

[0092] An upstream element may be formed from a material that is impermeable to air. In such embodiments, the aerosol-generating article may be configured such that air flows into the rod of aerosol-generating substrate through suitable ventilation means provided in a wrapper.

[0093] An upstream element may be formed of a solid cylindrical plug element having a filled cross-section. Such a plug element may be referred to as a 'plain' element. The solid plug element may be porous, as described above, but does not have a tubular form and therefore does not provide a longitudinal flow channel. The solid plug element preferably has a substantially uniform transverse cross section.

[0094] An upstream element may be formed of a hollow tubular segment defining a longitudinal cavity providing an unrestricted flow channel. As such, an upstream element can provide protection for the aerosol-generating substrate, as described above, whilst having a minimal effect on the overall resistance to draw (RTD) and filtration properties of the article.

[0095] Preferably, the diameter of the longitudinal cavity of the hollow tubular segment forming an upstream element is at least 3 millimetres, more preferably at least 3.5 millimetres, more preferably at least 4 millimetres and more preferably at least 4.5 millimetres. Preferably, the diameter of the longitudinal cavity is maximised in order to minimise the RTD of the upstream section, or an upstream element thereof.

[0096] Preferably, the wall thickness of the hollow tubular segment is less than 2 millimetres, more preferably less than 1.5 millimetres and more preferably less than 1 millimetre.

[0097] An upstream element of the upstream section may be made of any material suitable for use

in an aerosol-generating article. The upstream element may, for example, be made of a same material as used for one of the other components of the aerosol-generating article, such as the downstream filter segment or the hollow tubular cooling element. Suitable materials for forming the upstream element include filter materials, ceramic, polymer material, cellulose acetate, cardboard, zeolite, or aerosol-generating substrate. The upstream element may comprise a plug of cellulose acetate. The upstream element may comprise a hollow acetate tube, or a cardboard tube. [0098] Preferably, the upstream section, or an upstream element thereof, has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. Preferably, the external diameter of the upstream section, or an upstream element thereof, is between 5 millimetres and 8 millimetres, more preferably between 5.25 millimetres and 7.5 millimetres, more preferably between 5.5 millimetres and 7 millimetres.

[0099] Preferably, the upstream section or an upstream element has a length of between 2 millimetres and 10 millimetres, more preferably between 3 millimetres and 8 millimetres, more preferably between 2 millimetres and 6 millimetres. In a particularly preferred embodiment, the upstream section or an upstream element has a length of 5 millimetres. The length of the upstream section or an upstream element can advantageously be varied in order to provide the desired total length of the aerosol-generating article. For example, where it is desired to reduce the length of one of the other components of the aerosol-generating article, the length of the upstream section or an upstream element may be increased in order to maintain the same overall length of the article.

[0100] In addition, the length of the upstream section, or an upstream element thereof, can be used to control the position of the aerosol-generating article within the cavity of an aerosol-generating device, for articles which are intended to be externally heated. This can advantageously ensure that the position of the aerosol-generating substrate within the cavity can be optimised for heating and the position of any ventilation can also be optimised.

[0101] The upstream section is preferably circumscribed by a wrapper, such as a plug wrap. The wrapper circumscribing the upstream section is preferably a stiff plug wrap, for example, a plug wrap having a basis weight of at least 80 grams per square metre (gsm), or at least 100 gsm, or at least 110 gsm. This provides structural rigidity to the upstream section.

[0102] The upstream section is preferably connected to the rod of aerosol-generating substrate and optionally at least a part of the downstream section by means of an outer wrapper, as described herein.

[0103] Preferably, the aerosol-generating articles according to the present disclosure as described herein are adapted for use in electrically operated aerosol-generating systems in which the aerosol-generating substrate of the heated aerosol-generating article is heated by an electrical heat source. As described herein, electrically operated aerosol-generating systems comprising an inductive heating device may also comprise the aerosol-generating article having the aerosol-generating substrate and a susceptor in thermal proximity to the aerosol-generating substrate. The susceptor may be in direct contact with the aerosol-generating substrate and heat is transferred from the susceptor to the aerosol-generating substrate primarily by conduction. Examples of electrically operated aerosol-generating systems having inductive heating devices and aerosol-generating articles having susceptors are described in WO-A1-95/27411 and WO-A1-2015/177255.

[0104] The aerosol-generating device may extend between a distal end and a mouth end. The aerosol-generating device may comprise a body or housing. The body or housing of the aerosol-generating device may define a device cavity, or heating chamber, for removably receiving the aerosol-generating article at the mouth end of the device. The aerosol-generating device may comprise a heating element or heater for heating the aerosol-generating substrate when the aerosol-generating article is received within the device cavity.

[0105] In other words, an aerosol-generating device may comprise a heating chamber for receiving an aerosol-generating article and a heating element provided at or about the periphery of the heating chamber.

[0106] The device cavity may be referred to as the heating chamber of the aerosol-generating device. The device cavity may extend between a distal end and a mouth, or proximal, end. The distal end of the device cavity may be a closed end and the mouth, or proximal, end of the device cavity may be an open end. The mouth end or open end of the device cavity may correspond to the mouth end or distal end of the aerosol-generating device. The aerosol-generating device may be configured to receive an aerosol-generating article through the mouth end of the device or device cavity (or heating chamber). The aerosol-generating device may be configured to receive an aerosol-generating article via the mouth end of the device or device cavity (or heating chamber). The device cavity or heating chamber may be configured to receive an aerosol-generating article through or via its mouth end. An aerosol-generating article may be configured to be received into or within the device or device cavity (or heating chamber) through or via the mouth end of the device or device cavity. An aerosol-generating article may be configured to be inserted into the device or device cavity (or heating chamber) via or through the mouth end of the device or device cavity. An aerosol-generating article may be inserted into the device cavity, or heating chamber, via the open end of the device or device cavity. The device cavity may be cylindrical in shape so as to conform to the same shape of an aerosol-generating article.

[0107] The expression “received within” may refer to the fact that a component or element is fully or partially received within another component or element. For example, the expression “aerosol-generating article is received within the device cavity” refers to the aerosol-generating article being fully or partially received within the device cavity of the aerosol-generating article.

[0108] When the aerosol-generating article is received within the device cavity, the aerosol-generating article may abut the distal end of the device cavity. When the aerosol-generating article is received within the device cavity, the aerosol-generating article may be in substantial proximity to the distal end of the device cavity. The distal end of the device cavity may be defined by an end-wall.

[0109] When received within the device or device cavity (or heating chamber), the aerosol-generating article may be configured to protrude or extend beyond the mouth end of the aerosol-generating device. The length of the aerosol-generating article may be greater than the length of the device cavity (or heating chamber). This allows ease of insertion and removal of an article from the device and enables a mouth end portion of the article to extend beyond the device on which a user may draw aerosol.

[0110] The length of the device cavity may be between about 15 millimetres and about 80 millimetres. Preferably, the length of the device cavity is between about 20 millimetres and about 70 millimetres. More preferably, the length of the device cavity is between about 25 millimetres and about 60 millimetres. More preferably, the length of the device is between about 25 millimetres and about 50 millimetres.

[0111] The length of the device cavity may be between about 25 millimetres and about 29 millimetres. Preferably, the length of the device cavity is between about 25 millimetres and about 29 millimetres. More preferably, the length of the device cavity is between about 26 millimetres and about 29 millimetres. Even more preferably, the length of the device cavity is about 27 millimetres or about 28 millimetres.

[0112] A diameter of the device cavity may be between about 4 millimetres and about 10 millimetres. A diameter of the device cavity may be between about 5 millimetres and about 9 millimetres. A diameter of the device cavity may be between about 6 millimetres and about 8 millimetres. A diameter of the device cavity may be between about 6 millimetres and about 7.5 millimetres.

[0113] A diameter of the device cavity may be substantially the same as or greater than a diameter of the aerosol-generating article. A diameter of the device cavity may be the same as a diameter of the aerosol-generating article in order to establish a tight fit with the aerosol-generating article.

[0114] The device cavity may be configured to establish a tight fit with an aerosol-generating

article received within the device cavity. Tight fit may refer to a snug fit. The aerosol-generating device may comprise a peripheral wall. Such a peripheral wall may define the device cavity, or heating chamber. The peripheral wall defining the device cavity may be configured to engage with an aerosol-generating article received within the device cavity in a tight fit manner, so that there is substantially no gap or empty space between the peripheral wall defining the device cavity and the aerosol-generating article when received within the device.

[0115] Such a tight fit may establish an airtight fit or configuration between the device cavity and an aerosol-generating article received therein. With such an airtight configuration, there would be substantially no gap or empty space between the peripheral wall defining the device cavity and the aerosol-generating article for air to flow through. The tight fit with an aerosol-generating article may be established along the entire length of the device cavity or along a portion of the length of the device cavity.

[0116] The aerosol-generating device may comprise an air intake channel extending between a channel inlet and a channel outlet. The air intake channel may be configured to establish a fluid communication between the interior of the device cavity and the exterior of the aerosol-generating device. The air intake channel of the aerosol-generating device may be defined within the housing of the aerosol-generating device to enable fluid communication between the interior of the device cavity and the exterior of the aerosol-generating device. When an aerosol-generating article is received within the device cavity, the air intake channel may be configured to provide air flow into the article in order to deliver generated aerosol to a user drawing from the mouth end of the article.

[0117] The air intake channel of the aerosol-generating device may be defined within, or by, the peripheral wall of the housing of the aerosol-generating device. In other words, the air intake channel of the aerosol-generating device may be defined within the thickness of the peripheral wall or by the inner surface of the peripheral wall, or a combination of both. The air intake channel may partially be defined by the inner surface of the peripheral wall and may be partially defined within the thickness of the peripheral wall. The inner surface of the peripheral wall defines a peripheral boundary of the device cavity.

[0118] The air intake channel of the aerosol-generating device may extend from an inlet located at the mouth end, or proximal end, of the aerosol-generating device to an outlet located away from mouth end of the device. The air intake channel may extend along a direction parallel to the longitudinal axis of the aerosol-generating device.

[0119] The heating chamber or device cavity may be sized such that a longitudinal gap is provided between an aerosol-generating article received therein and the peripheral wall defining the device cavity. Such a longitudinal gap may partially or entirely circumscribe an aerosol-generating article received within the device. Such a longitudinal gap or space may define an air intake channel extending from the open, mouth end of the device cavity to the closed, distal end of the device cavity. Further, the device housing may be configured such that air may enter the upstream end of the article when the upstream end of the article abuts the distal end of the device cavity. The device housing and cavity may be such that a fluid communication between the air intake channel of the device, preferably at the distal end of the device cavity, and the upstream end of a received aerosol-generating article is established. As a result, upon drawing on the inserted aerosol-generating article, air may enter the aerosol-generating device through the air intake channel and flow towards the distal end of the device cavity and enter the upstream end of a received article.

[0120] The heater may be any suitable type of heater. Preferably, in the present disclosure, the heater is an external heater.

[0121] The heating element of such aerosol-generating devices may be of any suitable form to conduct heat. The heating of the aerosol-generating substrate may be achieved internally, externally, or both. The heating element may be a heater blade or pin adapted to be inserted into the aerosol-generating substrate so that the substrate is heated from inside. The heating element may preferably partially or completely surround the substrate and externally heat the substrate

circumferentially from the outside.

[0122] Preferably, the heater may externally heat the aerosol-generating article when received within the aerosol-generating device. Such an external heater may circumscribe or surround the aerosol-generating article when inserted in or received within the aerosol-generating device. Preferably, the length of the heater substantially corresponds to the length of the aerosol-generating substrate of an aerosol-generating article, which the aerosol-generating device is configured to receive.

[0123] The heater may be arranged to heat the outer surface of the aerosol-generating substrate. The heater may be arranged for insertion into an aerosol-generating substrate when the aerosol-generating substrate is received within the cavity. The heater may be positioned within the device cavity, or heating chamber.

[0124] The heater may comprise at least one heating element. The at least one heating element may be any suitable type of heating element. The device may comprise only one heating element. The device comprises a plurality of heating elements. The heater may comprise at least one resistive heating element. Preferably, the heater comprises a plurality of resistive heating elements. Preferably, the resistive heating elements are electrically connected in a parallel arrangement. Advantageously, providing a plurality of resistive heating elements electrically connected in a parallel arrangement may facilitate the delivery of a desired electrical power to the heater while reducing or minimising the voltage required to provide the desired electrical power.

Advantageously, reducing or minimising the voltage required to operate the heater may facilitate reducing or minimising the physical size of the power supply.

[0125] The heater may comprise an inductive heating arrangement. The inductive heating arrangement may comprise an induction source and a susceptor, which may be provided externally to the aerosol-generating substrate or internally within the aerosol-generating substrate. The induction source may comprise an inductor coil and a power supply configured to provide high frequency oscillating current to the inductor coil. As used herein, a high frequency oscillating current means an oscillating current having a frequency of between about 500 kHz and about 30 MHz. The heater may advantageously comprise a DC/AC inverter for converting a DC current supplied by a DC power supply to the alternating current. The inductor coil may be arranged to generate a high frequency oscillating electromagnetic field on receiving a high frequency oscillating current from the power supply. The inductor coil may be arranged to generate a high frequency oscillating electromagnetic field in the device cavity. The inductor coil may substantially circumscribe the device cavity. The inductor coil may extend at least partially along the length of the device cavity.

[0126] The heater may comprise an inductive heating element. The inductive heating element may be a susceptor element. As used herein, the term 'susceptor element' refers to an element comprising a material that is capable of converting electromagnetic energy into heat. When a susceptor element is located in an alternating electromagnetic field, the susceptor is heated. Heating of the susceptor element may be the result of at least one of hysteresis losses and eddy currents induced in the susceptor, depending on the electrical and magnetic properties of the susceptor material.

[0127] A susceptor element may be arranged such that, when the aerosol-generating article is received in the cavity of the aerosol-generating device, the oscillating electromagnetic field generated by the inductor coil induces a current in the susceptor element, causing the susceptor element to heat up. The aerosol-generating device is preferably capable of generating a fluctuating electromagnetic field having a magnetic field strength (H-field strength) of between 1 and 5 kilo amperes per metre (kA m), preferably between 2 and 3 kA/m, for example about 2.5 kA/m. The electrically operated aerosol-generating device is preferably capable of generating a fluctuating electromagnetic field having a frequency of between 1 and 30 MHz, for example between 1 and 10 MHz, for example between 5 and 7 MHz.

[0128] A susceptor element may be located in contact with the aerosol-generating substrate. A susceptor element may be located in the aerosol-generating device. A susceptor element may be located in or around the periphery of the device cavity. The aerosol-generating device may comprise only one susceptor element. The aerosol-generating device may comprise a plurality of susceptor elements. The susceptor element is preferably arranged to externally heat the aerosol-generating substrate. The susceptor element may circumscribe the aerosol-generating article when received within the heating chamber.

[0129] The susceptor element may comprise any suitable material. The susceptor element may be formed from any material that can be inductively heated to a temperature sufficient to release volatile compounds from the aerosol-generating substrate. Suitable materials for the elongate susceptor element include graphite, molybdenum, silicon carbide, stainless steels, niobium, aluminium, nickel, nickel containing compounds, titanium, and composites of metallic materials. Some susceptor elements comprise a metal or carbon. Advantageously the susceptor element may comprise or consist of a ferromagnetic material, for example, ferritic iron, a ferromagnetic alloy, such as ferromagnetic steel or stainless steel, ferromagnetic particles, and ferrite. A suitable susceptor element may be, or comprise, aluminium. The susceptor element preferably comprises more than about 5 percent, preferably more than 20 percent, more preferably more than 50 percent or more than 90 percent of ferromagnetic or paramagnetic materials. Some elongate susceptor elements may be heated to a temperature in excess of 250 degrees Celsius.

[0130] The susceptor element may comprise a non-metallic core with a metal layer disposed on the non-metallic core. For example, the susceptor element may comprise metallic tracks formed on an outer surface of a ceramic core or substrate.

[0131] The aerosol-generating device may comprise at least one resistive heating element and at least one inductive heating element. The aerosol-generating device may comprise a combination of resistive heating elements and inductive heating elements.

[0132] During use, the heater may be controlled to operate within a defined operating temperature range, below a maximum operating temperature. An operating temperature range between about 150 degrees Celsius and about 300 degrees Celsius in the heating chamber (or device cavity) is preferable. The operating temperature range of the heater may be between about 150 degrees Celsius and about 250 degrees Celsius.

[0133] Preferably, the operating temperature range of the heater may be between about 150 degrees Celsius and about 200 degrees Celsius. More preferably, the operating temperature range of the heater may be between about 180 degrees Celsius and about 200 degrees Celsius. In particular, it has been found that optimal and consistent aerosol delivery may be achieved when using an aerosol-generating device having an external heater, which has an operating temperature range between about 180 degrees Celsius and about 200 degrees Celsius, with aerosol-generating articles having a relatively low RTD (for example, with a downstream section RTD of less than 15 millimetres H.sub.2O), as mentioned in the present disclosure.

[0134] As described above, the hollow tubular substrate element of aerosol-generating articles according to the present disclosure can advantageously be adapted such that the length substantially matches the longitudinal dimensions of the heating element of the aerosol-generating system which is intended to be used to heat the aerosol-generating article. This may ensure that the hollow tubular substrate element is heated along substantially its full length, so that the generation of aerosol from the aerosol-generating substrate can be maximised.

[0135] The aerosol-generating device may comprise a power supply. The power supply may be a DC power supply. In some embodiments, the power supply is a battery. The power supply may be a nickel-metal hydride battery, a nickel cadmium battery, or a lithium-based battery, for example a lithium-cobalt, a lithium-iron-phosphate, or a lithium-polymer battery. However, in some embodiments the power supply may be another form of charge storage device, such as a capacitor. The power supply may require recharging and may have a capacity that allows for the storage of

enough energy for one or more user operations, for example one or more aerosol-generating experiences. For example, the power supply may have sufficient capacity to allow for continuous heating of an aerosol-generating substrate for a period of around six minutes, corresponding to the typical time taken to smoke a conventional cigarette, or for a period that is a multiple of six minutes. In another example, the power supply may have sufficient capacity to allow for a predetermined number of puffs or discrete activations of the heater.

[0136] As discussed above, the aerosol-generating device comprises an airflow guiding element extending along the device cavity, or heating chamber. The airflow guiding element preferably extends along the central longitudinal axis of the heating chamber.

[0137] An aerosol-generating article according to the present disclosure preferably comprises an open upstream end. Such an open upstream end may be defined by an open upstream end of the aerosol-generating substrate of the aerosol-generating article. Such an open upstream end may be defined by an open upstream end of the upstream section of the aerosol-generating article. For example, a hollow tubular element of the upstream section or a hollow tubular aerosol-generating substrate may define an open upstream end of the article. Further, the hollow tubular substrate and, if present, a hollow tubular upstream element define longitudinally extending cavities arranged to receive the airflow guiding element upon insertion of the aerosol-generating article into the heating chamber of the aerosol-generating device.

[0138] When an article according to the present disclosure is inserted into the heating chamber, the airflow guiding element preferably extends longitudinally into the longitudinal substrate cavity. A primary airflow path or channel may be defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate. A width or a diameter of a portion of the airflow guiding element may therefore be smaller than a diameter of the substrate cavity.

[0139] The airflow guiding element may longitudinally extend from the device housing into the heating chamber. The airflow guiding element may longitudinally extend from a portion of the device housing defining a distal end of the heating chamber. The airflow guiding element preferably extends in a longitudinal direction towards the mouth end of the heating chamber. The airflow guiding element preferably extends in a longitudinal direction substantially parallel to the central axis of the heating chamber. The airflow guiding element preferably extends in a longitudinal direction substantially aligned with the central axis of the heating chamber.

[0140] An external heater of the aerosol-generating device may partially or wholly surround the airflow guiding element extending along the heating chamber.

[0141] The aerosol-generating device may comprise a base support element, from which the airflow guiding element may extend. The base support element may be coupled to a distal end of the device cavity.

[0142] A maximum external diameter of the base support element may be approximately equal to the external diameter of the aerosol-generating substrate. A maximum external diameter of the base support element may be approximately equal to the internal diameter of the aerosol-generating substrate. A maximum external diameter of the base support element may be approximately equal to the internal diameter of the hollow tubular segment of an upstream section.

[0143] The base support element may have a disc or plate shape, preferably cylindrically shaped. Preferably, the base support element is porous or comprise at least one aperture. This enables a fluid communication to be established between the exterior of the aerosol-generating article and the interior of the aerosol-generating substrate through the base support element. The periphery of the base support element may be chamfered. This may facilitate engagement of the base support element with a hollow upstream end of an aerosol-generating article upon insertion into the device cavity. The chamfered periphery of the base support element may function as a centering guide for an article being inserted into the aerosol-generating device, ensuring that an inserted article is centered about the central longitudinal axis of the device cavity.

[0144] The airflow guiding element preferably comprises an elongate body extending longitudinally into the heating chamber. The airflow guiding element preferably comprises an elongate body extending from an upstream, proximal end of the airflow guiding element to a downstream, distal end of the airflow guiding element. The airflow guiding element preferably comprises an elongate body extending from a fixed end of the airflow guiding element to a free end of the airflow guiding element. The downstream or distal end of the airflow guiding element may be closer to the mouth or distal end of the aerosol-generating device than the upstream or proximal end of the airflow guiding element.

[0145] The upstream end of the airflow guiding element may be coupled to the base support element. A central longitudinal axis of the airflow guiding element may be aligned with a central longitudinal axis of the base support element. The airflow guiding element preferably comprises an elongate body extending from a fixed end of the airflow guiding element to a free end of the airflow guiding element. The end of the airflow guiding element coupled to the base support element may define the fixed end of the airflow guiding element, while the opposite end may be the free end of the airflow guiding element.

[0146] The length of the base support element may be at least about 0.5 mm. The length of the base support element may be at least about 1 mm. The length of the base support element may be at least about 1.5 mm.

[0147] The length of the base support element may be up to about 5 mm. The length of the base support element may be up to about 4 mm. The length of the base support element may be up to about 3 mm.

[0148] The elongate body of the airflow guiding element may be rod shaped or conically shaped. The elongate body of the airflow guiding element preferably comprises a hollow body or tube defining a longitudinal empty cavity. The elongate body of the airflow guiding element preferably comprises a hollow cylindrical tube defining a longitudinal empty cavity.

[0149] The downstream end of the hollow body may be closed, such that air may not flow into the substrate cavity via the interior of the hollow body.

[0150] The airflow guiding element may comprise an airflow inlet at a first position and an airflow outlet at a second position downstream of the first position, such that an airflow pathway is defined within and along the airflow guiding element.

[0151] The downstream end of the hollow body may be porous or may be provided with one or more perforations, apertures, inlets or outlets, such that fluid communication is established between the interior of the hollow body and the substrate cavity and a secondary airflow path into the substrate cavity is defined.

[0152] The downstream end of the hollow body may be open, such that fluid communication is established between the interior of the hollow body and the substrate cavity and a secondary airflow path within the airflow guiding element is defined. The secondary airflow path may direct air directly to the downstream section of the article if the airflow guiding element spans the entire length of the substrate cavity. The secondary airflow path may direct air into the substrate cavity of the article if the airflow guiding element extends along less than 100 percent of the length of the substrate cavity.

[0153] The upstream end of the hollow body may also be open. The base support element is preferably porous or preferably comprises at least one aperture such that fluid communication is established between the exterior of the aerosol-generating article and the interior of the aerosol-generating substrate via the base support element. The base support element may comprise a central aperture aligned with the open upstream end of the airflow guiding element. This allows air to flow through the base support element into the substrate cavity via a central secondary airflow path defined within the airflow guiding element. The primary airflow path or channel may be that defined between the interior surface of the aerosol-generating substrate and an exterior surface of the airflow guiding element. In other words, the primary airflow path or channel may circumscribe

the airflow guiding element.

[0154] Where the airflow guiding element comprises a hollow body, perforations or holes may be provided extending through the peripheral wall of the hollow body that defines a longitudinal empty cavity. The peripheral wall of the hollow body may be porous. Such one or more peripheral perforations may trace a particular path along or around the hollow body of the airflow guiding element. Such one or more peripheral perforations may trace a linear, a helicoidal, a curved, or a wavelike path along or around the hollow body of the airflow guiding element. This beneficially allows a fluid communication to be established between the interior of the hollow body and the substrate cavity such that one or more secondary airflow paths into the substrate cavity are defined. Air may flow through the base support element into the hollow body or tube of the airflow guiding element and exit into the substrate cavity through the peripheral wall of the hollow body. Air travelling through the hollow body may also exit into the substrate cavity via an opening at the downstream end of the hollow body. Such secondary airflows may provide a diluting and cooling function to the primary airflow travelling downstream between the airflow guiding element and the inner surface of the aerosol-generating element.

[0155] The width or diameter of the airflow guiding element may be uniform along its length. The width or diameter of the airflow guiding element may vary along its length. The inner diameter of the airflow guiding element may be at least about 0.5 mm. The inner diameter of the airflow guiding element may be at least about 1 mm. The inner diameter of the airflow guiding element may be at least about 1.5 mm. The inner diameter of the airflow guiding element may be at least about 2 mm. The inner diameter of the airflow guiding element may be up to about 5 mm. The inner diameter of the airflow guiding element may be up to about 4 mm. The size of the hollow cavity defined within the airflow guiding element defines the size of the secondary airflow channel or path and the amount of diluting or cooling air arranged to enter the substrate cavity during use.

[0156] The airflow guiding element may be textured. The airflow guiding element may have an uneven outer surface or raised outer surface. The airflow guiding element may comprise raised elements on its outer surface. The airflow guiding element may comprise one or more grooves, dimples, bumps, projections, ridges, or bulges provided on its outer surface. By having a textured or uneven external surface, the airflow guiding element is capable of disturbing the air flowing around it so as to form localised turbulence, which may enhance mixing of air with released aerosol-forming components.

[0157] The airflow guiding element may comprise a plurality or series of aligned segments forming an elongate body. Each of the segments may have any shape. The segments may each be pyramid, rod, cylindrically, conically, circular, spherically, or hemispherically shaped. For example, the airflow guiding element may comprise a plurality of aligned conically shaped segments to form an elongate body.

[0158] The airflow guiding element may comprise a core elongate portion, in accordance with the elongate body described above, and at least one extension portion located along the core portion. The at least one extension portion may comprise a bump, a projection, or a bulge provided on the outer surface of the core portion. The at least one extension portion may be formed on an external surface of the core portion. The at least one extension portion may define a raised surface on the airflow guiding element, preferably the core portion thereof. The at least one extension portion may extend outwardly (in other words, away from the central axis of the core portion) from the core portion. The at least one extension portion may extend radially from the core portion. The at least one extension portion may extend longitudinally along the core portion. The at least one extension portion may extend circumferentially, either entirely or partially, around the core portion.

[0159] The airflow guiding element may comprise at least two extension portions located along the core portion. Each extension portion may be located at respective longitudinal or axial positions along the core portion. The airflow guiding element may comprise three extension portions located along the core portion.

[0160] An extension portion may be substantially shaped in the form of a sphere, a hemisphere, a cylinder, or a ring. By providing one or more extension portions along the core portion of the airflow guiding element, the width or diameter of the airflow guiding element may vary along its length so as to encourage localised airflow separation from an external surface of the airflow guiding element. This in turn may encourage localised turbulent airflow to form between the airflow guiding element and an inner surface of the aerosol-generating substrate, thereby improving aerosol generation as a result of the enhanced mixing of the air with aerosol-forming components released from the heated aerosol-generating substrate.

[0161] Each extension portion of the airflow guiding element may extend along the core portion by a certain length. The length of each or an extension portion of the airflow guiding element may be at least about 5 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be at least about 10 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be at least about 15 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be at least about 20 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be at least about 25 percent of the total length of the airflow guiding element.

[0162] The length of each or an extension portion of the airflow guiding element may be up to about 75 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be up to about 60 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be up to about 50 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be up to about 40 percent of the total length of the airflow guiding element. The length of each or an extension portion of the airflow guiding element may be up to about 35 percent of the total length of the airflow guiding element.

[0163] Where a plurality of extension portions is provided, each extension portion may have a different length. For example, the airflow guiding element may comprise three extension portions arranged in sequential order along the core portion, the first extension portion may extend along about 15 percent of the total length of the airflow guiding element and the other two extension portions may extend along about 35 percent of the total length of the airflow guiding element.

[0164] Advantageously, the amount of disturbance and turbulence created in the air flowing around the airflow guiding element can be tailored, depending on the size of the extension portions relative to the airflow guiding element, the location of an or each extension portion along the airflow guiding element and the distance amongst or between sequential or adjacent extension portions.

[0165] An extension portion may be provided at an upstream end of the airflow guiding element or core portion thereof. An extension portion may be provided at a downstream end of the airflow guiding element or core portion thereof.

[0166] Sequential or adjacent extension portions may be spaced apart from each other so as to define a gap between them. The extension portions may be evenly spaced apart from each other. Providing a gap between sequential or adjacent extension portions defines a section of the primary airflow channel where the cross-sectional area is increased, allowing a local deceleration of the air after flowing over an upstream extension portion or before flowing over a downstream extension portion, mainly when the cross-section of the extension portion is uniform along its length like a cylinder or a ring. An outer surface of the core portion may be exposed by such a gap.

[0167] The base support element and the airflow guiding element may be manufactured separately and subsequently coupled to each other prior to assembly of the aerosol-generating device. The base support element and the airflow guiding element may be formed integral to each other, for example, by extrusion or by injection moulding. Similarly, the core portion and any extension

portion of the airflow guiding element may be manufactured separately and subsequently coupled to each other prior to assembly of the aerosol-generating device. The core portion and any extension portion of the airflow guiding element may be formed integral to each other, for example, by extrusion or by injection moulding. Extension portions may be manufactured separately to the core portion and may be subsequently assembled onto the core portion. For example, an extension portion may be a ring- or cylinder-shaped element that is mounted onto and coupled to the core portion. For example, such a ring- or cylinder-shaped element may be slid onto the core portion and engaged with the core portion by means of adhesion or an interference fit. An airflow guiding element construction that is substantially structurally resistant to deformation is preferable for use in an aerosol-generating device due its intended repeated use with the insertion and removal of aerosol-generating articles.

[0168] The material of the base support element and the material of the airflow guiding element may be the same. The airflow guiding element, the base support element, or both may be formed from a non-metallic material. The airflow guiding element, the base support element, or both may not comprise a metallic material. The airflow guiding element, the base support element, or both may be formed from cardboard. The airflow guiding element, the base support element, or both may be formed from a paper-based material. The airflow guiding element, the base support element, or both may be formed from paper. The airflow guiding element, the base support element, or both may be formed from a polymeric material. The airflow guiding element, the base support element, or both may be formed from a plastic material. The airflow guiding element, the base support element, or both may be formed from a bioplastic material.

[0169] The airflow guiding element, the base support element, or both may be formed from cellulose acetate. The materials mentioned in the present disclosure may provide suitable resistance to deformation or compression, while providing a base support element and an airflow guiding element that can be manufactured cost effectively.

[0170] The airflow guiding element, the base support element, or both may comprise a thermally conductive material. This may facilitate the heat transfer to the inner surface of the aerosol-generating substrate, particularly when heated by an external heating element.

[0171] The airflow guiding element may comprise an outer layer or coating at least partially provided on an external surface, preferably an external surface of the airflow guiding element body. The outer layer or coating may comprise one or more of an aerosol-former, a flavorant, and a further aerosol-generating substrate. Such aerosol-former, flavorant, and further aerosol-generating substrate of the outer layer or coating may respectively be in accordance with an aerosol-former, a flavorant, and the aerosol-generating substrate as described within the present disclosure.

[0172] The length of the airflow guiding element preferably corresponds to the amount by which the airflow guiding element extends into the substrate cavity.

[0173] The length of the airflow guiding element may be at least about 1 mm. The length of the airflow guiding element may be at least about 3 mm. The length of the airflow guiding element may be at least about 5 mm. The length of the airflow guiding element may be at least about 6 mm. The length of the airflow guiding element may be at least about 8 mm. The length of the airflow guiding element may be at least about 9 mm.

[0174] The length of the airflow guiding element may be up to about 30 mm. The length of the airflow guiding element may be up to about 25 mm. The length of the airflow guiding element may be up to about 20 mm. The length of the airflow guiding element may be up to about 15 mm. The length of the airflow guiding element may be up to about 12 mm. The length of the airflow guiding element may be up to about 10 mm.

[0175] The length of the airflow guiding element may be between about 1 mm and about 30 mm, between about 3 mm and about 30 mm, between about 5 mm and about 30 mm, between about 6 mm and about 30 mm, between about 8 mm and about 30 mm, or preferably between about 9 mm and about 30 mm. The length of the airflow guiding element may be between about 1 mm and

about 25 mm, between about 3 mm and about 25 mm, between about 5 mm and about 25 mm, between about 6 mm and about 25 mm, between about 8 mm and about 25 mm, or preferably between about 9 mm and about 25 mm. The length of the airflow guiding element may be between about 1 mm and about 20 mm, between about 3 mm and about 20 mm, between about 5 mm and about 20 mm, between about 6 mm and about 20 mm, between about 8 mm and about 20 mm, or preferably between about 9 mm and about 20 mm. The length of the airflow guiding element may be between about 1 mm and about 15 mm, between about 3 mm and about 15 mm, between about 5 mm and about 15 mm, between about 6 mm and about 15 mm, between about 8 mm and about 15 mm, or preferably between about 9 mm and about 15 mm. The length of the airflow guiding element may be between about 1 mm and about 12 mm, between about 3 mm and about 12 mm, between about 5 mm and about 12 mm, between about 6 mm and about 12 mm, between about 8 mm and about 12 mm, or preferably between about 9 mm and about 12 mm. The length of the airflow guiding element may be between about 1 mm and about 10 mm, between about 3 mm and about 10 mm, between about 5 mm and about 10 mm, between about 6 mm and about 10 mm, between about 8 mm and about 10 mm, or preferably between about 9 mm and about 10 mm.

[0176] The length of the airflow guiding element may be at least about 10 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 10 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 20 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 25 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 30 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 40 percent of the length of the diameter of the heating chamber (or device cavity). The length of the airflow guiding element may be at least about 50 percent of the length of the diameter of the heating chamber (or device cavity).

[0177] The length of the airflow guiding element defines the length of the restricted airflow path or channel defined between an external surface of the airflow guiding element and an interior surface of the aerosol-generating substrate. It may be realised that a balance can be struck between providing a relatively long airflow guiding element so as to define a relatively long restricted airflow path and the manufacturing costs of providing such a long airflow guiding element. Ideally, the length of the airflow guiding element may be at least about 50 percent of the length of the diameter of the heating chamber (or device cavity), preferably at least about 60 percent.

[0178] The length of the airflow guiding element may be at least about 10 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 25 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 30 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 50 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 60 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 75 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 80 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 90 percent of the length of the aerosol-generating substrate, or substrate cavity. The length of the airflow guiding element may be at least about 100 percent of the length of the aerosol-generating substrate, or substrate cavity.

[0179] The length of the airflow guiding element defines the length of the restricted airflow path or channel defined between an external surface of the airflow guiding element and an interior surface of the aerosol-generating substrate. It may be realised that a balance can be struck between

providing a relatively long airflow guiding element so as to define a relatively long restricted airflow path and the manufacturing costs of providing such a long airflow guiding element. Ideally, the length of the airflow guiding element may be at least about 50 percent of the length of the aerosol-generating substrate, or substrate cavity, preferably at least about 60 percent.

[0180] Preferably, the central or longitudinal axis of the airflow guiding element is aligned with the central or longitudinal axis of the aerosol-generating substrate. Preferably, the airflow guiding element is axially symmetric. This may ensure that the restricted airflow channel defined around the airflow guiding element and between the airflow guiding element and the internal surface of the aerosol-generating substrate is axially symmetric.

[0181] The airflow guiding element has a maximum external diameter. The one or more raised surfaces or extension portions of the airflow guiding element may define a maximum external diameter or width of the airflow guiding element. As such, the diameter or width of the airflow guiding element may vary or oscillate along the length of the airflow guiding element.

[0182] Where the external diameter or width of the airflow guiding element varies, the portion of the airflow guiding element having an external diameter or width corresponding to the maximum width or external diameter of the airflow guiding element is preferably located away from or downstream from the upstream end or proximal end of the airflow guiding element. This ensures that the air flowing around the airflow guiding element encounters a decrease in the cross-sectional area of the airflow channel defined between the external surface of the airflow guiding element and the internal surface of the aerosol-generating substrate received in the device cavity, as a result of the increase in diameter of the airflow guiding element, further along the airflow guiding element and not at the base (or upstream end or proximal end) of the airflow guiding element. This diameter or width increase can disturb the air flow and push the air closer to the internal surface of the aerosol-generating substrate, thereby encouraging localised turbulent airflow and localised air flow acceleration to form between the airflow guiding element, particularly at the location where the diameter or width increases, and an inner surface of the aerosol-generating substrate. This may improve aerosol generation as a result of the enhanced mixing of the air with aerosol-forming components released from the heated aerosol-generating substrate.

[0183] A maximum external diameter or width of the airflow guiding element may be at least about 1 mm. A maximum external diameter or width of the airflow guiding element may be at least about 2 mm. A maximum external diameter or width of the airflow guiding element may be at least about 2.5 mm. A maximum external diameter or width of the airflow guiding element may be at least about 3 mm.

[0184] A maximum external diameter or width of the airflow guiding element may be at least about 8 mm. A maximum external diameter or width of the airflow guiding element may be at least about 7.5 mm. A maximum external diameter or width of the airflow guiding element may be at least about 7 mm. A maximum external diameter or width of the airflow guiding element may be at least about 6 mm.

[0185] A maximum external diameter or width of the airflow guiding element may be between about 1 mm and about 8 mm, about 2 mm and about 8 mm, about 2.5 mm and about 8 mm, or preferably about 3 mm and about 8 mm. A maximum external diameter or width of the airflow guiding element may be between about 1 mm and about 7.5 mm, about 2 mm and about 7.5 mm, about 2.5 mm and about 7.5 mm, or preferably about 3 mm and about 7.5 mm. A maximum external diameter or width of the airflow guiding element may be between about 1 mm and about 7 mm, about 2 mm and about 7 mm, about 2.5 mm and about 7 mm, or preferably about 3 mm and about 7 mm. A maximum external diameter or width of the airflow guiding element may be between about 1 mm and about 6 mm, about 2 mm and about 6 mm, about 2.5 mm and about 6 mm, or preferably about 3 mm and about 6 mm.

[0186] The maximum external diameter or width of the airflow guiding element may be at least about 20 percent of the diameter of the heating chamber (or device cavity). The maximum external

diameter or width of the airflow guiding element may be at least about 35 percent of the diameter of the heating chamber (or device cavity). The maximum external diameter or width of the airflow guiding element may be at least about 40 percent of the diameter of the heating chamber (or device cavity). The maximum external diameter or width of the airflow guiding element may be at least about 50 percent of the diameter of the heating chamber (or device cavity). The maximum external diameter or width of the airflow guiding element may be at least about 60 percent of the diameter of the heating chamber (or device cavity).

[0187] The maximum external diameter or width of the airflow guiding element may be at least about 25 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The maximum external diameter or width of the airflow guiding element may be at least about 40 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The maximum external diameter or width of the airflow guiding element may be at least about 50 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The maximum external diameter or width of the airflow guiding element may be at least about 60 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The maximum external diameter or width of the airflow guiding element may be at least about 75 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity.

[0188] A portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element may be located away (or downstream) from the base or upstream end of the airflow guiding element. A portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element may be located at least about 10 percent of the length of the airflow guiding element away (or downstream) from the base or upstream end of the airflow guiding element. A portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element may be located at least about 20 percent of the length of the airflow guiding element away (or downstream) from the base or upstream end of the airflow guiding element. A portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element may be located at least about 25 percent of the length of the airflow guiding element away (or downstream) from the base or upstream end of the airflow guiding element. A portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element may be located at least about 50 percent of the length of the airflow guiding element away (or downstream) from the base or upstream end of the airflow guiding element.

[0189] The or any diameter of the portion of the airflow guiding element upstream of the portion of the airflow guiding element having a diameter corresponding to a maximum external diameter or width of the airflow guiding element is preferably less than (or does not exceed) the maximum external diameter or width of the airflow guiding element.

[0190] The airflow guiding element may have a minimum external diameter. For an airflow guiding element having a uniform external diameter or width, the diameter or width of the airflow guiding element may correspond to a maximum external diameter or width or a minimum external diameter or width. The core portion of the airflow guiding element may define a minimum external diameter or width of the airflow guiding element.

[0191] A minimum external diameter or width of the airflow guiding element may be at least about 0.5 mm. A minimum external diameter or width of the airflow guiding element may be at least about 1.5 mm. A minimum external diameter or width of the airflow guiding element may be at least about 2.5 mm. A minimum external diameter or width of the airflow guiding element may be at least about 3 mm.

[0192] A minimum external diameter or width of the airflow guiding element may be at least about 8 mm. A minimum external diameter or width of the airflow guiding element may be at least about

7.5 mm. A minimum external diameter or width of the airflow guiding element may be at least about 7 mm. A minimum external diameter or width of the airflow guiding element may be at least about 6 mm.

[0193] A minimum external diameter or width of the airflow guiding element may be between about 0.5 mm and about 8 mm, about 1.5 mm and about 8 mm, about 2.5 mm and about 8 mm, or preferably about 3 mm and about 8 mm. A minimum external diameter or width of the airflow guiding element may be between about 0.5 mm and about 7.5 mm, about 1.5 mm and about 7.5 mm, about 2.5 mm and about 7.5 mm, or preferably about 3 mm and about 7.5 mm. A minimum external diameter or width of the airflow guiding element may be between about 0.5 mm and about 7 mm, about 1.5 mm and about 7 mm, about 2.5 mm and about 7 mm, or preferably about 3 mm and about 7 mm. A minimum external diameter or width of the airflow guiding element may be between about 1 mm and about 6 mm, about 2 mm and about 6 mm, about 2.5 mm and about 6 mm, or preferably about 3 mm and about 6 mm.

[0194] The minimum external diameter or width of the airflow guiding element may be at least about 5 percent of the diameter of the heating chamber (or device cavity). The minimum external diameter or width of the airflow guiding element may be at least about 10 percent of the diameter of the heating chamber (or device cavity). The minimum external diameter or width of the airflow guiding element may be at least about 20 percent of the diameter of the heating chamber (or device cavity). The minimum external diameter or width of the airflow guiding element may be at least about 25 percent of the diameter of the heating chamber (or device cavity). The minimum external diameter or width of the airflow guiding element may be at least about 40 percent of the diameter of the heating chamber (or device cavity). The minimum external diameter or width of the airflow guiding element may be at least about 50 percent of the diameter of the heating chamber (or device cavity).

[0195] The minimum external diameter or width of the airflow guiding element may be at least about 10 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The minimum external diameter or width of the airflow guiding element may be at least about 20 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The minimum external diameter or width of the airflow guiding element may be at least about 25 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The minimum external diameter or width of the airflow guiding element may be at least about 40 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The minimum external diameter or width of the airflow guiding element may be at least about 50 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. The minimum external diameter or width of the airflow guiding element may be at least about 60 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity.

[0196] A maximum external diameter or width of the airflow guiding element may be about 100 percent of the internal diameter of the aerosol-generating substrate, or the diameter of the substrate cavity. As such, a portion of the airflow guiding element may be in contact with the aerosol-generating substrate. A portion of an extension or raised portion of the airflow guiding element may be in contact with the aerosol-generating substrate. Sizing the airflow guiding element to contact the aerosol-generating substrate may facilitate centering of the airflow guiding element with the substrate cavity and may assist in retaining airflow guiding element within the substrate cavity.

[0197] The airflow guiding element may extend along a central axis of the device cavity. The airflow guiding element may extend along a central axis of the substrate cavity. A longitudinal gap or space may be defined between an external surface of the airflow guiding element and the peripheral wall of the device defining the heating chamber.

[0198] When an article is received with the heating chamber, a longitudinal gap or space may be defined between an external surface of the airflow guiding element and the interior circumferential

surface of the aerosol-generating substrate. The external surface of the airflow guiding element, or a portion thereof, may be spaced from the interior surface of the substrate cavity such that an airflow channel is defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate. Preferably, such a defined airflow channel may be referred to as a primary airflow channel or restricted airflow channel.

[0199] The airflow channel is preferably annular in shape. In other words, the gap or space between the airflow guiding element and the internal surface of the aerosol-generating substrate may define a substantially annular chamber or channel.

[0200] A distance between an external surface of the airflow guiding element and the internal surface of the aerosol-generating substrate may define the height or thickness of the primary or restricted airflow channel. Subject to the local diameter or width of the airflow guiding element at a particular longitudinal and circumferential position along its elongate body, the height or thickness of the airflow channel may vary longitudinally or circumferentially or both.

[0201] A maximum thickness of the airflow channel may be at least about 0.25 mm. A maximum thickness of the airflow channel may be at least about 0.5 mm. A maximum thickness of the airflow channel may be at least about 1 mm. A maximum thickness of the airflow channel may be at least about 1.5 mm.

[0202] A minimum thickness of the airflow channel may be at least about 0 mm. In other words, a portion of the airflow guiding element may be in contact with the aerosol-generating substrate so that there is no distance or gap between such a portion of the airflow guiding element and the aerosol-generating substrate. A minimum thickness of the airflow channel may be at least about 0.25 mm. A minimum thickness of the airflow channel may be at least about 0.5 mm. A minimum thickness of the airflow channel may be at least about 1.5 mm. As described herein, upon receipt of the aerosol-generating article within the heating chamber of the device, the airflow guiding element extends longitudinally into the substrate cavity. In other words, the airflow guiding element is received within the aerosol-generating substrate of the aerosol-generating article. The airflow guiding element may not contact an internal surface of the aerosol-generating substrate. Along its entire length, the airflow guiding element may not contact an internal surface of the aerosol-generating substrate. In other words, no portion of the external surface of the airflow guiding element contacts an internal surface of the aerosol-generating substrate. This enables the definition of a thicker and less obstructed airflow channel, thereby enabling more airflow to travel between the internal surface of the aerosol-generating substrate and the airflow guiding element.

[0203] A maximum thickness of the airflow channel may be up to about 6 mm. A maximum thickness of the airflow channel may be up to about 5 mm. A maximum thickness of the airflow channel may be up to about 3 mm. A maximum thickness of the airflow channel may be up to about 2.5 mm.

[0204] A minimum thickness of the airflow channel may be up to about 6 mm. A minimum thickness of the airflow channel may be up to about 5 mm. A minimum thickness of the airflow channel may be up to about 3 mm. A minimum thickness of the airflow channel may be up to about 2.5 mm.

[0205] For an airflow guiding element having a substantially uniform diameter or width along its length, the thickness or height of the airflow channel may be uniform so that the minimum thickness and the maximum thickness of the airflow channel are effectively equivalent.

[0206] The airflow guiding element may be or may comprise a susceptor element configured to generate heat when placed in an oscillating electromagnetic field.

[0207] In the aerosol-generating articles of the present disclosure, the aerosol-generating substrate comprising the hollow tubular substrate element is combined with a downstream section, located downstream of the aerosol-generating substrate. The downstream section is preferably located immediately downstream of the aerosol-generating substrate. The downstream section of the aerosol-generating article preferably extends between the aerosol-generating substrate and the

downstream end of the aerosol-generating article. The downstream section may comprise one or more elements, each of which will be described in more detail within the present disclosure.

[0208] Preferably, the downstream section comprises at least one hollow tubular element. The hollow tubular element may be adjacent to the downstream end of the rod of aerosol-generating substrate. The hollow tubular element may be provided immediately downstream of the aerosol-generating substrate. In other words, the hollow tubular element may be at a downstream end of the aerosol-generating substrate. This arrangement may optimise flow of the aerosol from the longitudinal airflow channel of the hollow tubular substrate element into the downstream section and through the aerosol-generating article.

[0209] Preferably, the downstream section of the aerosol-generating article comprises a single hollow tubular element. In other words, the downstream section of the aerosol-generating article may comprise only one hollow tubular element.

[0210] The hollow tubular element of the downstream section may also be referred to as a hollow tubular downstream element.

[0211] In the context of the present disclosure, the hollow tubular element of the downstream section provides an unrestricted flow channel through the airflow passage. This means that the hollow tubular element provides a negligible level of resistance to draw (RTD), as defined above. The airflow passage should therefore be free from any components that would obstruct the flow of air in a longitudinal direction. Preferably, the airflow passage is substantially empty.

[0212] The hollow tubular element of the downstream section provides an empty cavity downstream of the aerosol-generating substrate, which may enhance cooling and nucleation of aerosol particles generated by the aerosol-generating substrate. The hollow tubular element of the downstream section therefore may function as an aerosol-cooling element.

[0213] The length of the hollow tubular element may be at least about 12 mm. The length of the hollow tubular element may be at least about 15 mm. The length of the hollow tubular element may be at least about 20 mm.

[0214] The length of the hollow tubular element of the downstream section may be less than or equal to about 50 mm. The length of the hollow tubular element may be less than or equal to about 45 mm. The length of the hollow tubular element may be less than or equal to about 40 mm.

[0215] For example, the length of the hollow tubular element of the downstream section may be between about 12 mm and 50 mm. The length of the hollow tubular element may be between about 15 mm and 45 mm. The length of the hollow tubular element may be between about 20 mm and 40 mm. The length of the hollow tubular element may be about 30 mm.

[0216] A relatively long hollow tubular element provides and defines a relatively long internal cavity within the downstream section of the aerosol-generating article. Providing a relatively long cavity may maximise the nucleation benefits described above, thereby improving aerosol formation and cooling.

[0217] The ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be less than or equal to about 1.25. Preferably, a ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be less than or equal to about 1. More preferably, a ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be less than or equal to about 0.75.

[0218] The ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be at least about 0.2. Preferably, a ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be at least about 0.25. More preferably, a ratio between the length of the hollow tubular substrate element and the length of the hollow tubular element of the downstream section may be at least about 0.3.

[0219] For example, the ratio between the length of the hollow tubular substrate element and the

length of the hollow tubular element of the downstream section may be between about 0.2 and about 1.25, or between about 0.25 and about 1, or between about 0.3 and about 0.75.

[0220] The ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be less than or equal to about 1. Preferably, the ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be less than or equal to about 0.90. More preferably, the ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be less than or equal to about 0.85.

[0221] The ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be at least about 0.35. Preferably, the ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be at least about 0.45. More preferably, the ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be at least about 0.50.

[0222] For example, the ratio between the length of the hollow tubular element of the downstream section and the total length of the downstream section may be between about 0.35 and about 1, or between about 0.45 and about 0.9, or between about 0.5 and about 0.85.

[0223] The ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be less than or equal to about 0.80.

Preferably, the ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be less than or equal to about 0.70. More preferably, the ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be less than or equal to about 0.60.

[0224] The ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be at least about 0.25. Preferably, the ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be at least about 0.30. More preferably, the ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be at least about 0.40.

[0225] For example, the ratio between the length of the hollow tubular element of the downstream section and the overall length of the aerosol-generating article may be between about 0.25 and about 0.8, or between about 0.3 and about 0.7, or between about 0.4 and about 0.6.

[0226] The wall thickness of the hollow tubular element of the downstream section may be at least about 100 micrometres. The wall thickness of the hollow tubular element of the downstream section may be at least about 150 micrometres. The wall thickness of the hollow tubular element of the downstream section may be at least about 200 micrometres, preferably at least about 250 micrometres and even more preferably at least about 500 micrometres (or 0.5 mm).

[0227] The wall thickness of the hollow tubular element of the downstream section may be less than or equal to about 2 millimetres, preferably less than or equal to about 1.5 millimetres and even more preferably less than or equal to about 1.25 mm. The wall thickness of the hollow tubular element of the downstream section may be less than or equal to about 1 millimetre. The wall thickness of the hollow tubular element of the downstream section may be less than or equal to about 500 micrometres.

[0228] The wall thickness of the hollow tubular element of the downstream section may be between about 100 micrometres and about 2 millimetres, preferably between about 150 micrometres and about 1.5 millimetres, even more preferably between about 200 micrometres and about 1.25 millimetres.

[0229] Keeping the wall thickness of the hollow tubular segment of the downstream section relatively low ensures that the overall internal volume of the hollow tubular element—which is

made available for the aerosol to begin the nucleation process as soon as the aerosol components leave the aerosol-generating substrate—and the cross-sectional surface area of the cavity of the hollow tubular element are effectively maximised, whilst at the same time ensuring that the hollow tubular element has the necessary structural strength to prevent a collapse of the aerosol-generating article as well as to provide some support to the rod of aerosol-generating substrate, and that the RTD of the hollow tubular element is minimised. Greater values of cross-sectional surface area of the cavity of the hollow tubular element are understood to be associated with a reduced speed of the aerosol stream travelling along the aerosol-generating article, which is also expected to favour aerosol nucleation. Further, it would appear that by utilising a hollow tubular element having a relatively low thickness, it is possible to substantially prevent diffusion of ventilation air prior to its contacting and mixing with the stream of aerosol, which is also understood to further favour nucleation phenomena. In practice, by providing a more controllably localised cooling of the stream of volatilised species, it may be possible to enhance the effect of cooling on the formation of new aerosol particles.

[0230] The hollow tubular element of the downstream section preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating substrate and to the external diameter of the aerosol-generating article. The hollow tubular element of the downstream section preferably has an external diameter that is greater than the external diameter of the hollow tubular substrate element of the aerosol-generating substrate.

[0231] The hollow tubular element may have an external diameter of between 5 millimetres and 10 millimetres, for example of between 5.5 millimetres and 9 millimetres or of between 6 millimetres and 8 millimetres.

[0232] The hollow tubular element of the downstream section may have a constant internal diameter along a length of the hollow tubular element. However, the internal diameter of the hollow tubular element may vary along the length of the hollow tubular element.

[0233] The hollow tubular element of the downstream section may have an internal diameter of at least about 2 millimetres. For example, the hollow tubular element may have an internal diameter of at least about 2.5 millimetres, at least about 3 millimetres, or at least about 3.5 millimetres. The provision of a hollow tubular element having an internal diameter as set out above may advantageously provide sufficient rigidity and strength to the hollow tubular element.

[0234] The hollow tubular element of the downstream section may have an internal diameter of no more than about 10 millimetres. For example, the hollow tubular element may have an internal diameter of no more than about 9 millimetres, no more than about 8 millimetres, or no more than about 7.5 millimetres. The provision of a hollow tubular element having an internal diameter as set out above may advantageously reduce the resistance to draw of the hollow tubular segment.

[0235] For example, the hollow tubular element of the downstream section may have an internal diameter of between about 2 millimetres and about 10 millimetres, between about 2.5 millimetres and about 9 millimetres, between about 3 millimetres and about 8 millimetres, or between about 3.5 millimetres and about 7.5 millimetres.

[0236] The ratio of the internal diameter of the hollow tubular substrate element to the internal diameter of the hollow tubular element of the downstream section is preferably between about 0.8 and about 1.2, more preferably between about 0.9 and about 1.1, most preferably about 1.

[0237] Particularly preferably, the internal diameter of the hollow tubular substrate element is substantially equal to the internal diameter of the hollow tubular element of the downstream section.

[0238] The central longitudinal axis of the hollow tubular substrate element of the aerosol-generating substrate may preferably be aligned with the central longitudinal axis of the hollow tubular element of the downstream section. For example, where the internal diameter of the hollow tubular substrate element is substantially equal to the internal diameter of the hollow tubular element of the downstream section, the central longitudinal axis of the hollow tubular substrate

element may be aligned with the central longitudinal axis of the hollow tubular substrate element of the downstream section so that the cavity of the hollow tubular substrate element and the cavity of the hollow tubular element of the downstream section may be substantially aligned.

[0239] The hollow tubular element of the downstream section may comprise a paper-based material. The hollow tubular element may comprise at least one layer of paper. The paper may be very rigid paper. The paper may be crimped paper, such as crimped heat resistant paper or crimped parchment paper.

[0240] Preferably, the hollow tubular element may comprise cardboard. The hollow tubular element may be a cardboard tube. The hollow tubular element may be formed from cardboard. Advantageously, cardboard is a cost-effective material that provides a balance between being deformable in order to provide ease of insertion of the aerosol-generating article into an aerosol-generating device and being sufficiently stiff to provide suitable engagement of the article with the interior of the device. A cardboard tube may therefore provide suitable resistance to deformation or compression during use.

[0241] The hollow tubular element of the downstream section may be a paper tube. The hollow tubular element may be a tube formed from spirally wound paper. The hollow tubular element may be formed from a plurality of layers of the paper. The paper may have a basis weight of at least about 50 grams per square meter, at least about 60 grams per square meter, at least about 70 grams per square meter, or at least about 90 grams per square meter.

[0242] The hollow tubular element of the downstream section may comprise a polymeric material. For example, the hollow tubular element may comprise a polymeric film. The polymeric film may comprise a cellulosic film. The hollow tubular segment may comprise low density polyethylene (LDPE) or polyhydroxyalkanoate (PHA) fibres. The hollow tubular element may comprise cellulose acetate tow.

[0243] Where the hollow tubular element comprises cellulose acetate tow, the cellulose acetate tow may have a denier per filament of between about 2 and about 4 and a total denier of between about 25 and about 40.

[0244] The hollow tubular element may be at the upstream end of the downstream section. The hollow tubular element may abut the downstream end of the aerosol-generating substrate. The hollow tubular element may abut the downstream end of the hollow tubular substrate element.

[0245] The aerosol-generating article according to the present disclosure may comprise a ventilation zone at a location along the downstream section. In more detail, where the downstream section comprises a hollow tubular element, the ventilation zone may be provided at a location along the hollow tubular element.

[0246] As such, a ventilated cavity is provided downstream of the rod of aerosol-generating substrate. This may provide particularly efficient cooling of the aerosol and promote enhanced nucleation of aerosol particles.

[0247] The ventilation zone may typically comprise a plurality of perforations through the peripheral wall of the hollow tubular element. The plurality of perforations of the ventilation zone may also be through any wrapper circumscribing the hollow tubular element. Preferably, the ventilation zone comprises at least one circumferential row of perforations. The ventilation zone may comprise two circumferential rows of perforations. For example, the perforations may be formed online during manufacturing of the aerosol-generating article. Preferably, each circumferential row of perforations comprises from **8 to 30** perforations.

[0248] The downstream section may further comprise a mouthpiece element. The mouthpiece element may be located at the downstream end of the aerosol-generating article. The mouthpiece element is preferably located downstream of the hollow tubular element of the downstream section, which is described above. The mouthpiece element may extend between the hollow tubular element of the downstream section and the downstream end of the aerosol-generating article.

[0249] The provision of a mouthpiece element at the downstream end of the aerosol-generating

articles according to the present disclosure may provide an appealing appearance and mouthfeel to the consumer.

[0250] The mouthpiece element may be a mouthpiece filter element. The mouthpiece element may comprise at least one mouthpiece filter segment formed of a fibrous filtration material. Parameters or characteristics described in relation to the mouthpiece element as a whole may equally be applied to a mouthpiece filter segment of the mouthpiece element.

[0251] The fibrous filtration material may be for filtering the aerosol that is generated from the aerosol-generating substrate. Suitable fibrous filtration materials would be known to the skilled person. Particularly preferably, the at least one mouthpiece filter segment comprises a cellulose acetate filter segment formed of cellulose acetate tow.

[0252] The mouthpiece element may consist of a single mouthpiece filter segment. The mouthpiece element may include two or more mouthpiece filter segments axially aligned in an abutting end to end relationship with each other.

[0253] The downstream section may comprise a mouth end cavity at the downstream end, downstream of the mouthpiece element as described above. The mouth end cavity may be defined by a further hollow tubular element provided at the downstream end of the mouthpiece element. The mouth end cavity may be defined by an outer wrapper of the aerosol-generating article, wherein the outer wrapper extends in a downstream direction from (or past) the mouthpiece element. For example, the mouth end cavity may be defined by a tipping wrapper extending downstream past the mouthpiece element.

[0254] The mouthpiece element may optionally comprise a flavourant, which may be provided in any suitable form. For example, the mouthpiece element may comprise one or more capsules, beads, or granules of a flavourant, or one or more flavour loaded threads or filaments.

[0255] Preferably, the mouthpiece element, or mouthpiece filter segment thereof, has a low particulate filtration efficiency.

[0256] Preferably, the mouthpiece element is circumscribed by a plug wrap. Preferably, the mouthpiece element is unventilated such that air does not enter the aerosol-generating article along the mouthpiece element.

[0257] The mouthpiece element preferably has an external diameter that is approximately equal to the external diameter of the aerosol-generating article. The diameter of a mouthpiece element (or mouthpiece filter segment) may be substantially the same as the external diameter of the hollow tubular element. As mentioned in the present disclosure, the external diameter of the hollow tubular element may be about 7.2 mm, plus or minus 10 percent.

[0258] The diameter of the mouthpiece element may be between about 5 mm and about 10 mm. The diameter of the mouthpiece element may be between about 5.5 mm and about 9 mm. The diameter of the mouthpiece element may be between about 6 mm and about 8 mm. The diameter of the mouthpiece element may be about 7.2 mm, plus or minus 10 percent. The diameter of the mouthpiece element may be about 7.25 mm, plus or minus 10 percent.

[0259] Unless otherwise specified, the resistance to draw (RTD) of a component or the aerosol-generating article is measured in accordance with ISO 6565-2015. The RTD refers to the pressure required to force air through the full length of a component. The terms “pressure drop” or “draw resistance” of a component or article may also refer to the “resistance to draw.”

[0260] The resistance to draw (RTD) of the downstream section may be at least about 0 mm H₂O. The RTD of the downstream section may be at least about 3 mm H₂O. The RTD of the downstream section may be at least about 6 mm H₂O.

[0261] The RTD of the downstream section may be no greater than about 12 mm H₂O. The RTD of the downstream section may be no greater than about 11 mm H₂O. The RTD of the downstream section may be no greater than about 10 mm H₂O.

[0262] The resistance to draw of the downstream section may be greater than or equal to about 0 mm H₂O and less than about 12 mm H₂O. Preferably, the resistance to draw of the downstream

section may be greater than or equal to about 3 mm H₂O and less than about 12 mm H₂O. The resistance to draw of the downstream section may be greater than or equal to about 0 mm H₂O and less than about 11 mm H₂O. Even more preferably, the resistance to draw of the downstream section may be greater than or equal to about 3 mm H₂O and less than about 11 mm H₂O. Even more preferably, the resistance to draw of the downstream section may be greater than or equal to about 6 mm H₂O and less than about 10 mm H₂O. Preferably, the resistance to draw of the downstream section may be about 8 mm H₂O.

[0263] The resistance to draw (RTD) characteristics of the downstream section may be wholly or mostly attributed to the RTD characteristics of the mouthpiece element of the downstream section. In other words, the RTD of the mouthpiece element of the downstream section may wholly define the RTD of the downstream section.

[0264] The resistance to draw (RTD) of the mouthpiece element may be at least about 0 mm H₂O. The RTD of the mouthpiece element may be at least about 3 mm H₂O. The RTD of the mouthpiece element may be at least about 6 mm H₂O.

[0265] The RTD of the mouthpiece element may be no greater than about 12 mm H₂O. The RTD of the mouthpiece element may be no greater than about 11 mm H₂O. The RTD of the mouthpiece element may be no greater than about 10 mm H₂O.

[0266] The resistance to draw of the mouthpiece element may be greater than or equal to about 0 mm H₂O and less than about 12 mm H₂O. Preferably, the resistance to draw of the mouthpiece element may be greater than or equal to about 3 mm H₂O and less than about 12 mm H₂O. The resistance to draw of the mouthpiece element may be greater than or equal to about 0 mm H₂O and less than about 11 mm H₂O. Even more preferably, the resistance to draw of the mouthpiece element may be greater than or equal to about 3 mm H₂O and less than about 11 mm H₂O. Even more preferably, the resistance to draw of the mouthpiece element may be greater than or equal to about 6 mm H₂O and less than about 10 mm H₂O. Preferably, the resistance to draw of the mouthpiece element may be about 8 mm H₂O.

[0267] As mentioned above, the mouthpiece element, or mouthpiece filter segment, may be formed of a fibrous material. The mouthpiece element may be formed of a porous material. The mouthpiece element may be formed of a biodegradable material. The mouthpiece element may be formed of a cellulose material, such as cellulose acetate. For example, a mouthpiece element may be formed from a bundle of cellulose acetate fibres having a denier per filament between about 10 and about 15. For example, a mouthpiece element formed from relatively low-density cellulose acetate tow, such as cellulose acetate tow comprising fibres of about 12 denier per filament.

[0268] The mouthpiece element may be formed of a polylactic acid-based material. The mouthpiece element may be formed of a bioplastic material, preferably a starch-based bioplastic material. The mouthpiece element may be made by injection moulding or by extrusion. Bioplastic-based materials are advantageous because they are able to provide mouthpiece element structures which are simple and cheap to manufacture with a particular and complex cross-sectional profile, which may comprise a plurality of relatively large air flow channels extending through the mouthpiece element material, which provides suitable RTD characteristics.

[0269] The mouthpiece element may be formed from a sheet of suitable material that has been crimped, pleated, gathered, woven, or folded into an element that defines a plurality of longitudinally extending channels. Such sheet of suitable material may be formed of paper, cardboard, a polymer, such as polylactic acid, or any other cellulose-based, paper-based material or bioplastic-based material. A cross-sectional profile of such a mouthpiece element may show the channels as being randomly oriented.

[0270] The mouthpiece element may be formed in any other suitable manner. For example, the mouthpiece element may be formed from a bundle of longitudinally extending tubes. The longitudinally extending tubes may be formed from polylactic acid. The mouthpiece element may be formed by extrusion, moulding, lamination, injection, or shredding of a suitable material. Thus,

it is preferred that there is a low-pressure drop (or RTD) from an upstream end of the mouthpiece element to a downstream end of the mouthpiece element.

[0271] The length of the mouthpiece element may be at least about 1.5 mm. The length of the mouthpiece element may be at least about 2 mm. The length of the mouthpiece element may equal to or less than about 7 mm. The length of the mouthpiece element may be equal to or less than about 4 mm. For example, the length of the mouthpiece element may be between about 1.5 mm and about 7 mm. The length of the mouthpiece element may be between about 2 millimetres and about 4 millimetres.

[0272] The ratio between the length of the mouthpiece element and the length of the downstream section may be less than or equal to about 0.35. Preferably, the ratio between the length of the mouthpiece element and the length of the downstream section may be less than or equal to about 0.30. More preferably, the ratio between a length of the mouthpiece element and the length of the downstream section may be less than or equal to about 0.25.

[0273] The ratio between the length of the mouthpiece element and the length of the downstream section may be at least about 0.03. Preferably, the ratio between a length of the mouthpiece element and the length of the downstream section may be at least about 0.05. More preferably, the ratio between the length of the mouthpiece element and the length of the downstream section may be at least about 0.1.

[0274] For example, the ratio between the length of the mouthpiece element and the length of the downstream section is from about 0.03 to about 0.35, preferably from about 0.05 to about 0.30, more preferably from about 0.1 to about 0.25.

[0275] The ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article may be less than or equal to about 0.20. Preferably, the ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article may be less than or equal to about 0.15. More preferably, the ratio between a length of the mouthpiece element and the overall length of the aerosol-generating article may be less than or equal to about 0.1.

[0276] The ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article may be at least about 0.01. Preferably, the ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article may be at least about 0.02. More preferably, the ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article may be at least about 0.05.

[0277] For example, the ratio between the length of the mouthpiece element and the overall length of the aerosol-generating article is from about 0.01 to about 0.2, preferably from about 0.02 to about 0.15, more preferably from about 0.05 to about 0.1.

[0278] Where the downstream section comprises a hollow tubular element and a mouthpiece element, a ratio of the length of the hollow tubular element to the length of the mouthpiece element may be at least about 1.5. In other words, the length of the hollow tubular element may be at least about 150% of the length of the mouthpiece element. The ratio of the length of the hollow tubular element to the length of the mouthpiece element may be at least about 5. The ratio of the length of the hollow tubular element to the length of the mouthpiece element may be at least about 7.5.

[0279] The ratio of the length of the hollow tubular element to the length of the mouthpiece element may be equal to or less than about 20. The ratio of the length of the hollow tubular element to the length of the mouthpiece element may be equal to or less than about 15. The ratio of the length of the hollow tubular element to the length of the mouthpiece element may be equal to or less than about 12.5.

[0280] For example, the ratio of the length of the hollow tubular element to the length of the mouthpiece element may be between about 1.5 and about 20, or between about 5 and about 15, or between about 7.5 and about 10.

[0281] The overall length of the downstream section is preferably at least about 15 millimetres,

more preferably at least about 20 millimetres, more preferably at least about 25 millimetres.

[0282] The overall length of the downstream section is preferably less than about 50 millimetres, more preferably less than about 45 millimetres, more preferably less than about 40 millimetres.

[0283] For example, the downstream section may have an overall length of between about 20 millimetres and about 50 millimetres, more preferably between about 25 millimetres and about 45 millimetres, more preferably between about 30 millimetres and about 40 millimetres.

[0284] The ratio between the total length of the downstream section and an overall length of the aerosol-generating article may be less than or equal to about 0.80. Preferably, the ratio between the length of the downstream section and an overall length of the aerosol-generating article may be less than or equal to about 0.75. More preferably, the ratio between the length of the downstream section and the overall length of the aerosol-generating article may be less than or equal to about 0.70. Even more preferably, the ratio between the length of the downstream section and the overall length of the aerosol-generating article may be less than or equal to about 0.65.

[0285] The ratio between the length of the downstream section and the overall length of the aerosol-generating article may be at least about 0.30. Preferably, the ratio between the length of the downstream section and the overall length of the aerosol-generating article may be at least about 0.40. More preferably, the ratio between a length of the downstream section and the overall length of the aerosol-generating article may be at least about 0.50. Even more preferably, the ratio between a length of the downstream section and the overall length of the aerosol-generating article may be at least about 0.60.

[0286] Preferably, an overall length of an aerosol-generating article in accordance with the invention is at least about 35 millimetres. More preferably, an overall length of an aerosol-generating article in accordance with the invention is at least about 40 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is at least about 45 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is at least about 50 millimetres.

[0287] An overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 110 millimetres. More preferably, an overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 100 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 75 millimetres. Even more preferably, an overall length of an aerosol-generating article in accordance with the invention is preferably less than or equal to 70 millimetres.

[0288] For example, the overall length of the aerosol-generating article may be between about 35 millimetres and about 110 millimetres, or between about 40 millimetres and about 100 millimetres, or between about 45 millimetres and about 75 millimetres, or between about 50 millimetres and about 70 millimetres.

[0289] The aerosol-generating article preferably has an external diameter of at least about 5 millimetres. Preferably, the aerosol-generating article has an external diameter of at least 5.5 millimetres. More preferably, the aerosol-generating article has an external diameter of at least 6 millimetres.

[0290] Preferably, the aerosol-generating article has an external diameter of less than or equal to about 10 millimetres. More preferably, the aerosol-generating article has an external diameter of less than or equal to about 9 millimetres. Even more preferably, the aerosol-generating article has an external diameter of less than or equal to about 8 millimetres.

[0291] For example, the aerosol-generating article may have an external diameter of between about 5 millimetres and about 10 millimetres, or between about 5.5 millimetres and about 9 millimetres, or between about 6 millimetres and about 8 millimetres.

[0292] The external diameter of the aerosol-generating article may be substantially constant over the whole length of the article. As an alternative, different portions of the aerosol-generating article

may have different external diameters.

[0293] One or more of the components of the aerosol-generating article may be individually circumscribed by their own wrapper.

[0294] Preferably, the aerosol-generating substrate and the downstream section are combined together with a wrapper, such as a tipping wrapper.

[0295] Preferably, the components of the aerosol-generating article according to the present disclosure are made from biodegradable materials.

[0296] Below, there is provided a non-exhaustive list of non-limiting examples. Any one or more of the features of these examples may be combined with any one or more features of another example, or embodiment, or aspect described herein.

[0297] EX1. An aerosol-generating device having a mouth end for receiving an aerosol-generating article, the aerosol-generating device comprising: [0298] a heating chamber for receiving an aerosol-generating article at the mouth end of the device.

[0299] EX2. An aerosol-generating device according to example EX1, further comprising an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the device, the airflow guiding element being configured to be inserted into an aerosol-generating article upon receipt within the heating chamber.

[0300] EX3. An aerosol-generating device according to example EX1 or EX2, further comprising a heating element provided at or about the periphery of the heating chamber for heating an aerosol-generating article received within the heating chamber.

[0301] EX4. An aerosol-generating device according to example EX2, wherein the airflow guiding element is a heating element for heating an aerosol-generating article received within the heating chamber.

[0302] EX5. An aerosol-generating device according to example EX2, wherein the airflow guiding element is not a heating element.

[0303] EX6. An aerosol-generating device according to example EX2, wherein the airflow guiding element is or comprises a susceptor element configured to generate heat when placed in an oscillating electromagnetic field.

[0304] EX7. An aerosol-generating device according to any preceding example, wherein the width or external diameter of the airflow guiding element varies along its length, wherein a first portion of the airflow guiding element has a width or external diameter corresponding to the maximum width or external diameter of the airflow guiding element, and a second portion of the airflow guiding element located upstream of said first portion has a width or external diameter smaller than the maximum width or external diameter of the airflow guiding element.

[0305] EX8. An aerosol-generating device according to any one of examples EX2 to EX7, wherein the width or diameter of the airflow guiding element varies along its length.

[0306] EX9. An aerosol-generating device according to any one of examples EX2 to EX8, wherein the airflow guiding element extends along at least 25 percent of the length of the heating chamber.

[0307] EX10. An aerosol-generating device according to any one of examples EX2 to EX9, wherein the airflow guiding element extends along at least 50 percent of the length of the heating chamber.

[0308] EX11. An aerosol-generating device according to any one of examples EX2 to EX10, wherein the airflow guiding element extends along at least 60 percent of the length of the heating chamber.

[0309] EX12. An aerosol-generating device according to any one of examples EX2 to EX11, wherein the airflow guiding element comprises a central core portion and an extension portion located along the core portion, wherein the extension portion extends outwardly from the core portion.

[0310] EX13. An aerosol-generating device according to example EX12, wherein the extension portion is substantially shaped in the form of a hemisphere, a sphere, a cylinder, a cone, or a ring.

[0311] EX14. An aerosol-generating device according to example EX12 or EX13, wherein the core portion is substantially shaped in the form of a rod, a tube, or a cone.

[0312] EX15. An aerosol-generating device according to any one of examples EX12 to EX14, wherein the airflow guiding element comprises at least two extension portions located along the core portion.

[0313] EX16. An aerosol-generating device according to any one of examples EX12 to EX14, wherein the airflow guiding element comprises at least two extension portions each located at different positions along the core portion.

[0314] EX17. An aerosol-generating device according to any one of examples EX2 to EX16, wherein the airflow guiding element comprises a hollow tube.

[0315] EX18. An aerosol-generating device according to any one of examples EX2 to EX17, further comprising a base support element and wherein the airflow guiding element extends from the base support element.

[0316] EX19. An aerosol-generating device according to example EX18, wherein the base support element is located at the upstream end of the heating chamber.

[0317] EX20. An aerosol-generating article according to example EX18 or EX19, wherein the base support element is porous or comprises at least one aperture such that a fluid communication is established through the base support element.

[0318] EX21. An aerosol-generating article according to any one of examples EX2 to EX20, wherein the airflow guiding element comprises an airflow inlet at a first position and an airflow outlet at a second position downstream of the first position, such that an airflow pathway is defined within and along the airflow guiding element.

[0319] EX22. An aerosol-generating article according to any one of examples EX2 to EX21, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the heating chamber is at least about 25 percent of the diameter of the heating chamber.

[0320] EX23. An aerosol-generating article according to any one of examples EX2 to EX22, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the heating chamber is at least about 50 percent of the diameter of the heating chamber.

[0321] EX24. An aerosol-generating article according to any one of examples EX2 to EX23, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the heating chamber is at least about 75 percent of the diameter of the heating chamber.

[0322] EX25. An aerosol-generating article according to any one of examples EX2 to EX24, wherein an outer surface of the airflow guiding element is textured.

[0323] EX26. An aerosol-generating system comprising an aerosol-generating device according to any one of the preceding examples and an aerosol-generating article configured to be received within the heating chamber.

[0324] EX27. An aerosol-generating system comprising an aerosol-generating device according to any one of the examples EX2 to EX25 and an aerosol-generating article configured to be received within the heating chamber, the aerosol-generating article comprising an aerosol-generating substrate, wherein the aerosol-generating substrate is in the form of a hollow tubular segment defining a substrate cavity extending from an upstream end of the aerosol-generating substrate to the downstream end of the aerosol-generating substrate, wherein upon receipt of the aerosol-generating article within the heating chamber of the device, the airflow guiding element extends longitudinally into the substrate cavity.

[0325] EX28. An aerosol-generating system according to example EX27, wherein a width or diameter of the airflow guiding element is smaller than a diameter of the substrate cavity.

[0326] EX29. An aerosol-generating system according to example EX27 or EX28, wherein the aerosol-generating article further comprises an upstream section located upstream of the aerosol-generating substrate, wherein the upstream section comprises an upstream element adjacent to the upstream end of the aerosol-generating substrate.

[0327] EX30. An aerosol-generating system according to any one of examples EX27 to EX29, wherein a portion of the airflow guiding element contacts a portion of the aerosol-generating substrate.

[0328] EX31. An aerosol-generating system according to any one of examples EX27 to EX30, wherein a portion of the airflow guiding element has a width or diameter that substantially corresponds to a diameter of the substrate cavity.

[0329] EX32. An aerosol-generating system according to any one of examples EX27 to EX31, wherein an outer longitudinal surface of the airflow guiding element and an inner longitudinal surface of the aerosol-generating substrate define the airflow channel.

[0330] EX33. An aerosol-generating system according to example EX29, wherein the upstream element comprises a solid plug segment.

[0331] EX34. An aerosol-generating system according to example EX29, wherein the upstream element comprises a hollow tubular segment.

[0332] EX35. An aerosol-generating system according to any one of examples EX27 to EX34, further comprising a downstream section located downstream of the aerosol-generating substrate, the downstream section comprising one or more of a mouthpiece element and a hollow tubular element.

[0333] EX36. An aerosol-generating article according to example EX27, wherein the mouthpiece element comprises at least one mouthpiece filter segment formed of a fibrous filtration material.

[0334] EX37. An aerosol-generating system according to any one of examples EX27 to EX36, wherein the aerosol-generating substrate has a length of between 5 millimetres and 30 millimetres.

[0335] EX38. An aerosol-generating system according to any one of examples EX27 to EX36, wherein the aerosol-generating substrate has a length of between 5 millimetres and 16 millimetres.

[0336] EX39. An aerosol-generating system according to any one of examples EX27 to EX38, wherein a wall thickness of the aerosol-generating substrate is between 5 percent and 40 percent of the external diameter of the aerosol-generating substrate.

[0337] EX40. An aerosol-generating system according to any one of examples EX27 to EX39, wherein a wall thickness of the aerosol-generating substrate is at least 200 micrometres.

[0338] EX41. An aerosol-generating system according to any one of examples EX27 to EX40, wherein the aerosol-generating substrate comprises homogenised tobacco material.

[0339] EX42. An aerosol-generating system according to any one of examples EX27 to EX41, wherein the aerosol-generating substrate is formed from a plurality of overlapping sheets of homogenised tobacco material.

[0340] EX43. An aerosol-generating system according to any one of examples EX27 to EX42, wherein the aerosol-generating substrate comprises one or more aerosol formers and wherein the content of aerosol former in the aerosol-generating substrate is between 10 percent and 20 percent by weight, on a dry weight basis.

[0341] EX44. An aerosol-generating system according to any one of examples EX27 to EX43, wherein a portion of the airflow guiding element is coated with one or more of a further aerosol-generating substrate, a flavourant, and an aerosol former.

[0342] EX45. An aerosol-generating system according to example EX43 or EX44, wherein the aerosol former comprises one or more of glycerine and propylene glycol.

[0343] EX46. An aerosol-generating system according to any one of examples EX27 to EX45, wherein the airflow guiding element extends along at least 25 percent of the length of the substrate cavity.

[0344] EX47. An aerosol-generating system according to any one of examples EX27 to EX45, wherein the airflow guiding element extends along at least 50 percent of the length of the substrate cavity.

[0345] EX48. An aerosol-generating system according to any one of examples EX27 to EX45, wherein the airflow guiding element extends along at least 60 percent of the length of the substrate

cavity.

[0346] EX49. An aerosol-generating system according to any one of examples EX27 to EX48, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the substrate cavity is at least about 25 percent of the diameter of the substrate cavity.

[0347] EX50. An aerosol-generating system according to any one of examples EX27 to EX48, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the substrate cavity is at least about 50 percent of the diameter of the substrate cavity.

[0348] EX51. An aerosol-generating system according to any one of examples EX27 to EX48, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the substrate cavity is at least about 75 percent of the diameter of the substrate cavity.

[0349] The present invention will be further described, by way of example only, with reference to the accompanying drawings in which:

Description

[0350] FIG. 1 shows a schematic side-sectional view of an aerosol-generating device in accordance with an embodiment of the present invention;

[0351] FIG. 2 shows a schematic side-sectional view of an aerosol-generating device in accordance with an embodiment of the present invention;

[0352] FIG. 3 shows a schematic side-sectional view of an aerosol-generating device in accordance with an embodiment of the present invention;

[0353] FIG. 4 shows a schematic side-sectional view of an aerosol-generating device in accordance with an embodiment of the present invention;

[0354] FIG. 5 shows a schematic side-sectional view of an aerosol-generating article for use in an aerosol-generating device in accordance with an embodiment of the present invention;

[0355] FIG. 6 shows a schematic side-sectional view of an aerosol-generating article for use in an aerosol-generating device in accordance with an embodiment of the present invention;

[0356] FIG. 7 shows a schematic side-sectional view of an aerosol-generating system comprising an aerosol-generating article, and an aerosol-generating device in accordance with an embodiment of the present invention;

[0357] FIG. 8 shows a schematic side-sectional view of an aerosol-generating system comprising an aerosol-generating article, and an aerosol-generating device in accordance with an embodiment of the present invention;

[0358] FIG. 9 shows a schematic side-sectional view of an aerosol-generating device in accordance with an embodiment of the present invention; and

[0359] FIG. 10 shows a schematic side-sectional view of an aerosol-generating system comprising an aerosol-generating article, and an aerosol-generating device in accordance with an embodiment of the present invention;

[0360] FIG. 1 shows a mouth end portion of an aerosol-generating device **100** configured to receive and heat a corresponding aerosol-generating article. In particular, FIG. 1 shows a downstream, mouth end portion of the aerosol-generating device **100** where a device cavity (or heating chamber) **108** of the device is defined and an aerosol-generating article can be received. A compatible aerosol-generating article may comprise a hollow, open distal end and a hollow aerosol-generating element.

[0361] The aerosol-generating device **100** comprises a housing (or body) **104**, extending between a mouth end **102** and a distal end (not shown). The housing **104** comprises a peripheral wall **106**. The peripheral wall **106** defines a device cavity or heating chamber **108** for receiving an aerosol-generating article. The device cavity **108** is defined by a closed, distal end **103** and an open, mouth end **102**. The mouth end of the device cavity **108** is located at the mouth end **102** of the aerosol-

generating device **100**. An aerosol-generating article is configured to be received through the mouth end of the device cavity **108** and retained within the device cavity **108**.

[0362] The aerosol-generating device **100** further comprises a heater **110** and a power source (not shown) for supplying power to the heater **110**. A controller (not shown) is also provided to control such supply of power to the heater. The heater **110** is configured to controllably heat an aerosol-generating article during use, when an aerosol-generating article is received within the device **100**. In the present embodiments, the heater **110** is arranged to externally heat the aerosol-generating substrate **12** of an aerosol-generating article during use.

[0363] The aerosol-generating device **100** further comprises an airflow guiding element **20** extending along the device cavity **108** from the closed, distal end **103**. The airflow guiding element **20** extends along the central longitudinal axis of the device cavity **108**. The airflow guiding element **20** has a fixed, upstream end and a free, downstream end.

[0364] The airflow guiding element **20** comprises an irregular external surface such that the external diameter of the airflow guiding element **20** varies along its length.

[0365] The airflow guiding element **202** comprises a core elongate portion **21** and a plurality of extension portions **23**, **27** extending radially outwardly from the core elongate portion **21**. The airflow guiding element **202** comprises three extension portions **23**, **27**. A first extension portion **27** is located at the upstream, fixed end of the airflow guiding element **20** and is hemispherically shaped. The upstream end of the first extension portion **27** is flat and is coupled to the distal end **104** of the device cavity **108**. As shown in FIG. **1**, downstream of the first extension portion **231** are two sequentially arranged, spherical extension portions **23**. One of the extension portions **23** is positioned at the downstream, free end of the airflow guiding element **20** and the other is positioned shortly upstream thereof, in between an extension portion **27** and the other extension portion **23**.

[0366] The airflow guiding element **20** is made from or comprises cardboard.

[0367] The length of the airflow guiding element **20** is about 8 mm. A maximum external diameter or width of the airflow guiding element **20** is about 3 mm. A minimum external diameter or width of the airflow guiding element **20** is about 1 mm. The diameter of the device cavity **108** is between about 7.25 mm and about 7.3 mm. The length of the device cavity **108** is at least about 15 mm.

[0368] FIG. **2** shows an aerosol-generating device **200** differing from the aerosol-generating device **100** shown in FIG. **1** in that the airflow guiding element **201** has a different structure, the device **200** comprises a base support element **25** and a longer device cavity **208**. The airflow guiding element **201** comprises an irregular external surface such that the external diameter of the airflow guiding element **201** varies along its length. The airflow guiding element **201** comprises a core elongate portion **21** and a plurality of extension portions **23** extending radially outwardly from the core elongate portion **21**. The airflow guiding element **201** comprises three extension portions **23**, the first one being positioned at the downstream, free end of the airflow guiding element **201**, a second one being positioned shortly upstream of the first extension portion **23** and a third one being positioned at the upstream, fixed end of the airflow guiding element **201**. The extension portions **23** are spherically shaped. The airflow guiding element **201** is made from or comprises cardboard.

[0369] The aerosol-generating device **200** comprises a base support element **25** coupled to the distal, closed end of the device cavity **103**. The airflow guiding element **201** is coupled to the downstream end of the base support element **25** and extends downstream therefrom. The base support element **25** comprises a porous material or one or more holes (not shown) extending through its length so as to allow air to flow through it when an aerosol-generating article is received within the device cavity **208**. The length of the base support element **25** is about 3 mm. In other words, the base support element **25** projects from the closed end **103** of the device cavity **208** by about 3 mm.

[0370] The periphery **26** of the base support element **25** is chamfered to facilitate engagement with the hollow upstream end of an aerosol-generating article upon insertion into the device cavity **208**.

The chamfered periphery **26** of the base support element **25** functions as a centering guide for an article being inserted into the aerosol-generating device **200**, ensuring that an inserted article is centered about the central longitudinal axis of the device cavity **208**. The maximum external diameter or width of the base support element **25** preferably matches the internal diameter of the hollow upstream section of an aerosol-generating article.

[0371] The length of the airflow guiding element **201** is about 10 mm. A maximum external diameter or width of the airflow guiding element **201** is about 3 mm. A minimum external diameter or width of the airflow guiding element **201** is about 1 mm. The diameter of the device cavity **208** is between about 7.25 mm and about 7.3 mm. The length of the device cavity **208** is at least about 20 mm.

[0372] FIG. **3** shows an aerosol-generating device **300** differing from the aerosol-generating device **100** shown in FIG. **1** in that the airflow guiding element **202** has a different structure. The airflow guiding element **202** comprises an elongate body in the form of a hollow tube having a uniform external diameter and a closed downstream end. The length of the airflow guiding element **202** is about 10 mm. The external diameter of the airflow guiding element **202** is about 3 mm. The airflow guiding element **202** is made from or comprises cardboard.

[0373] FIG. **4** shows an aerosol-generating device **400** differing from the aerosol-generating device **100** shown in FIG. **1** in that the airflow guiding element **203** has a different structure. The airflow guiding element **203** comprises an irregular external surface such that the external diameter of the airflow guiding element **203** varies along its length. The airflow guiding element **203** comprises a core portion **213** and a plurality of extension portions **233** extending radially outwardly from the core portion **213**. The airflow guiding element **203** comprises four extension portions **234**. The four extension portions **234** are cylindrically shaped protrusions evenly spaced along the core portion **213**. In this embodiment, the most upstream and downstream extension portions **234** are respectively spaced from the upstream and downstream ends of the airflow guiding element **203**.

[0374] The airflow guiding element **203** is made from or comprises cardboard.

[0375] The length of the airflow guiding element **203** is about 8 mm. A maximum external diameter or width of the airflow guiding element **203** is about 3 mm. A minimum external diameter or width of the airflow guiding element **203** is about 1 mm.

[0376] An aerosol-generating article **1** shown in FIG. **5** comprises a rod of aerosol-generating substrate **12** and a downstream section **14** provided downstream of the rod of aerosol-generating substrate **12**. The aerosol-generating article **1** extends from an upstream or distal end **16**—which coincides with an upstream end of the aerosol-generating substrate **12**—to a downstream or mouth end **18**, which coincides with a downstream end of the downstream section **14**. The downstream section **14** may comprise one or more of a hollow tubular element or a mouthpiece element (not shown), as described within the present disclosure.

[0377] The aerosol-generating article **1** has an outer diameter of about 7.25 mm.

[0378] The aerosol-generating substrate **12** comprises a hollow tubular substrate element **40** formed of homogenised tobacco material. The hollow tubular substrate element **40** has a peripheral wall **42** which defines a longitudinal cavity **44** providing an unrestricted flow channel through the hollow tubular substrate element **40**. The upstream end of the longitudinal cavity **44** provides an air inlet through which air can be drawn into the aerosol-generating article **10** during use. The hollow tubular substrate element **40** has a length of about 12 millimetres and an external diameter of about 7.25 mm. The wall thickness of the hollow tubular substrate element **40** is about 1 mm and the diameter of the substrate cavity **44** is 5.25 mm.

[0379] Each of the components of the aerosol-generating articles shown in the figures and described in the present disclosure may be circumscribed by corresponding wrappers or may be joined together by one or more wrappers, which are not shown in the figures.

[0380] FIG. **6** shows an aerosol-generating article **2** differing from the aerosol-generating article **1** in that it also comprises an upstream section **30** provided upstream of the aerosol-generating

substrate **12**. The upstream section **30** comprises an upstream element **34** in the form of a hollow tubular element defining an empty longitudinal cavity extending along its entire length. The downstream end of the upstream element **34** abuts the upstream end of the aerosol-generating substrate **12**. The upstream end **16** is defined by the upstream end of the upstream element **34**. The length of the hollow upstream element **34** is about 5 mm. The external and internal diameters of the upstream element **34** correspond to those of the aerosol-generating element **12** located immediately downstream.

[0381] FIG. 7 illustrates an exemplary aerosol-generating system **10** comprising the aerosol-generating device **100** and the aerosol-generating article **1** received within the device cavity (or heating chamber) **108** of the aerosol-generating device **100**. Upon insertion of the aerosol-generating article **1** into the device cavity **108**, the airflow guiding element **20** is received within the substrate cavity **44** of the article **1** and the hollow aerosol-generating substrate **12** of the article **1** effectively surrounds the airflow guiding element **20**. The aerosol-generating device **100** is configured so that, during use, air is configured to enter the device cavity **108** and into an aerosol-generating article via its upstream end when drawn on its downstream end.

[0382] An annular airflow channel **22** is defined between the internal surface of the hollow aerosol-generating substrate **12** and the external surface of the airflow guiding element **20**. Upon a user drawing on the mouth end **18** of the aerosol-generating article **1**, air can be drawn into the device **100** and around the article **1** towards the distal, closed end **103** of the device cavity **108**, where the air can reverse its direction and enter the article **1** via its upstream end. The air can then flow through the annular airflow channel **22** and progress towards to the downstream end **18** of the article **1**.

[0383] A maximum thickness of the airflow channel **22** is about 1.13 mm and a minimum thickness of the airflow channel **22** is about 2.13 mm.

[0384] FIG. 8 illustrates an exemplary aerosol-generating system **101** comprising the aerosol-generating device **200** and the aerosol-generating article **2** received within the device cavity (or heating chamber) **208** of the aerosol-generating device **200**. Upon insertion of the aerosol-generating article **2** into the device cavity **208**, the airflow guiding element **201** is received within the substrate cavity **44** of the article **2** and the hollow aerosol-generating substrate **12** of the article **2** effectively surrounds the airflow guiding element **201**. The hollow upstream end of the article **2** engages with the base support element **25**. The aerosol-generating device **200** is configured so that, during use, air is configured to enter the device cavity **208** and into an aerosol-generating article via its upstream end when drawn on its downstream end.

[0385] An annular airflow channel **22** is defined between the internal surface of the hollow aerosol-generating substrate **12** and the external surface of the airflow guiding element **201**. Upon a user drawing on the mouth end **18** of the aerosol-generating article **2**, air can be drawn into the device **200** and around the article **2** towards the distal, closed end **103** of the device cavity **108**, where the air can reverse its direction and enter the article **2** via its upstream end through the base support element **25**. The air can then flow through the annular airflow channel **22** and progress towards to the downstream end **18** of the article **1**.

[0386] A maximum thickness of the airflow channel **22** is about 1.13 mm and a minimum thickness of the airflow channel **22** is about 2.13 mm.

[0387] FIG. 9 shows an aerosol-generating device **500** differing from the aerosol-generating device **100** shown in FIG. 1 in that the airflow guiding element **204** has a different structure. The airflow guiding element **204** comprises an elongate body in the form of a hollow tube having a uniform external diameter and an open downstream end. The length of the airflow guiding element **204** is about 10 mm. The external diameter of the airflow guiding element **20** is about 3 mm. The airflow guiding element **204** is made from or comprises cardboard. The internal diameter of the airflow guiding element **204** is about 1.5 mm.

[0388] FIG. 10 illustrates an exemplary aerosol-generating system **102** comprising the aerosol-

generating device **500** and the aerosol-generating article **1** received within the device cavity (or heating chamber) **108** of the aerosol-generating device **500**. Upon insertion of the aerosol-generating article **1** into the device cavity **108**, the airflow guiding element **204** is received within the substrate cavity **44** of the article **1** and the hollow aerosol-generating substrate **12** of the article **1** effectively surrounds the airflow guiding element **201**. The hollow upstream end of the article **1** engages with the base support element **25**. Similarly, an annular airflow channel **22** is defined between the internal surface of the hollow aerosol-generating substrate **12** and the external surface of the airflow guiding element **204**. Upon a user drawing on the mouth end **18** of the aerosol-generating article **1**, air can be drawn into the device **200** and around the article **1** towards the distal, closed end **103** of the device cavity **108**, where the air can reverse its direction and enter the article **2** via its upstream end. The air can then flow through the annular airflow channel **22** and progress towards to the downstream end **18** of the article **1**. In addition to air flowing around the airflow guiding element **204** along the annular airflow channel **22**, air can flow into the interior of the hollow tube of the airflow guiding element **204** through a porous portion (not shown) present at the base of the airflow guiding element **204**. The internal longitudinal cavity defined by the airflow guiding element **204** defines a secondary airflow channel **24** extending from the upstream end of the airflow guiding element **204** and exiting via the open downstream end of the airflow guiding element **204**.

[0389] A thickness of the airflow channel **22** is about 2.13 mm. The internal diameter of the airflow guiding element **204** shown defines the diameter or width of the secondary airflow channel **24**.

[0390] In all figures of the present disclosure, air flow paths, aerosol flow paths, or other fluid paths into and through the aerosol-generating articles and devices during use are depicted with discontinuous arrows.

[0391] For the purpose of the present description and of the appended claims, except where otherwise indicated, all numbers expressing amounts, quantities, percentages, and so forth, are to be understood as being modified in all instances by the term “about”. Also, all ranges include the maximum and minimum points disclosed and include any intermediate ranges therein, which may or may not be specifically enumerated herein. In this context, therefore, a number A is understood as $A \pm 10\%$ of A. Within this context, a number A may be considered to include numerical values that are within general standard error for the measurement of the property that the number A modifies. The number A, in some instances as used in the appended claims, may deviate by the percentages enumerated above provided that the amount by which A deviates does not materially affect the basic and novel characteristic(s) of the claimed invention. Also, all ranges include the maximum and minimum points disclosed and include any intermediate ranges therein, which may or may not be specifically enumerated herein.

[0392] The specific embodiments and examples described above illustrate but do not limit the present invention. It is to be understood that other embodiments of the present invention may be made, and the specific embodiments and examples described herein are not exhaustive.

Claims

1.-15. (canceled)

16. An aerosol-generating system, comprising: an aerosol-generating device having a mouth end configured to receive an aerosol-generating article; and an aerosol-generating article configured to produce an inhalable aerosol upon heating, wherein the aerosol-generating device comprises: a heating chamber configured to receive and to heat the aerosol-generating article, and an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the aerosol-generating device, the airflow guiding element being configured to be inserted into the aerosol-generating article upon receipt within the heating chamber, wherein a width or external diameter of the airflow guiding element varies along a length of the airflow

guiding element, and wherein a first portion of the airflow guiding element has a width or external diameter corresponding to a maximum width or external diameter of the airflow guiding element, and a second portion of the airflow guiding element located upstream of the first portion has a width or external diameter smaller than the maximum width or external diameter of the airflow guiding element, wherein the aerosol-generating article comprises: an aerosol-generating substrate in a form of a hollow tubular segment defining a substrate cavity extending from an upstream end of the aerosol-generating substrate to a downstream end of the aerosol-generating substrate, and a downstream section located downstream of the aerosol-generating substrate, the downstream section comprising one or more of a mouthpiece element and a hollow tubular element, wherein upon receipt of the aerosol-generating article within the heating chamber of the aerosol-generating device, the airflow guiding element extends longitudinally into the substrate cavity, and wherein an airflow channel is defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate.

17. The aerosol-generating system according to claim 16, wherein the aerosol-generating device further comprises a heating element configured to heat the aerosol-generating article received within the heating chamber.

18. The aerosol-generating system according to claim 17, wherein the heating element is provided at or about a periphery of the heating chamber.

19. An aerosol-generating system, comprising: an aerosol-generating device having a mouth end configured to receive an aerosol-generating article; and an aerosol-generating article configured to produce an inhalable aerosol upon heating, wherein the aerosol-generating device comprises: a heating chamber configured to receive the aerosol-generating article at the mouth end of the aerosol-generating device, a heating element provided at or about a periphery of the heating chamber and being configured to heat the aerosol-generating article received within the heating chamber, and an airflow guiding element located within the heating chamber and extending longitudinally towards the mouth end of the device, the airflow guiding element being configured to be inserted into the aerosol-generating article upon receipt within the heating chamber, wherein the aerosol-generating article comprises: an aerosol-generating substrate in a form of a hollow tubular segment defining a substrate cavity extending from an upstream end of the aerosol-generating substrate to the downstream end of the aerosol-generating substrate, wherein, upon receipt of the aerosol-generating article within the heating chamber of the device, the airflow guiding element extends longitudinally into the substrate cavity, and wherein an airflow channel is defined between an external surface of the airflow guiding element and an internal surface of the aerosol-generating substrate, and a downstream section located downstream of the aerosol-generating substrate, the downstream section comprising one or more of a mouthpiece element and a hollow tubular element.

20. The aerosol-generating system according to claim 16, wherein an outer surface of the airflow guiding element is textured or uneven.

21. The aerosol-generating system according to claim 16, wherein a maximum width or diameter of a portion of the airflow guiding element extending into the heating chamber is at least about 25 percent of a diameter of the heating chamber.

22. The aerosol-generating system according to claim 16, wherein a maximum width or diameter of a portion of the airflow guiding element extending into the heating chamber is at least about 50 percent of a diameter of the heating chamber.

23. The aerosol-generating system according to claim 16, wherein the airflow guiding element extends along at least 50 percent of a length of the heating chamber.

24. The aerosol-generating system according to claim 16, wherein the airflow guiding element comprises an airflow inlet at a first position and an airflow outlet at a second position downstream of the first position, such that an airflow pathway is defined within and along the airflow guiding element.

- 25.** The aerosol-generating system according to claim 16, wherein the airflow guiding element comprises a hollow tube.
- 26.** The aerosol-generating system according to claim 16, wherein the airflow guiding element comprises an elongate body comprising a central core portion and an extension portion located along the central core portion, and wherein the extension portion extends outwardly from the central core portion.
- 27.** The aerosol-generating system according to claim 26, wherein the airflow guiding element comprises at least two extension portions located along the central core portion.
- 28.** The aerosol-generating system according to claim 16, wherein a wall thickness of the aerosol-generating substrate is between 5 percent and 40 percent of an external diameter of the aerosol-generating substrate.
- 29.** The aerosol-generating system according to claim 16, wherein a width or diameter of the airflow guiding element is smaller than a diameter of the substrate cavity.
- 30.** The aerosol-generating system according to claim 16, wherein a maximum width or diameter of the portion of the airflow guiding element extending into the substrate cavity is at least about 25 percent of a diameter of the substrate cavity.
- 31.** The aerosol-generating system according to claim 16, wherein the aerosol-generating article, when received within the aerosol-generating device, is further configured to extend beyond the mouth end of the aerosol-generating device.
- 32.** The aerosol-generating system according to claim 16, wherein the aerosol-generating article is further configured to be received through a mouth end of the heating chamber.
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